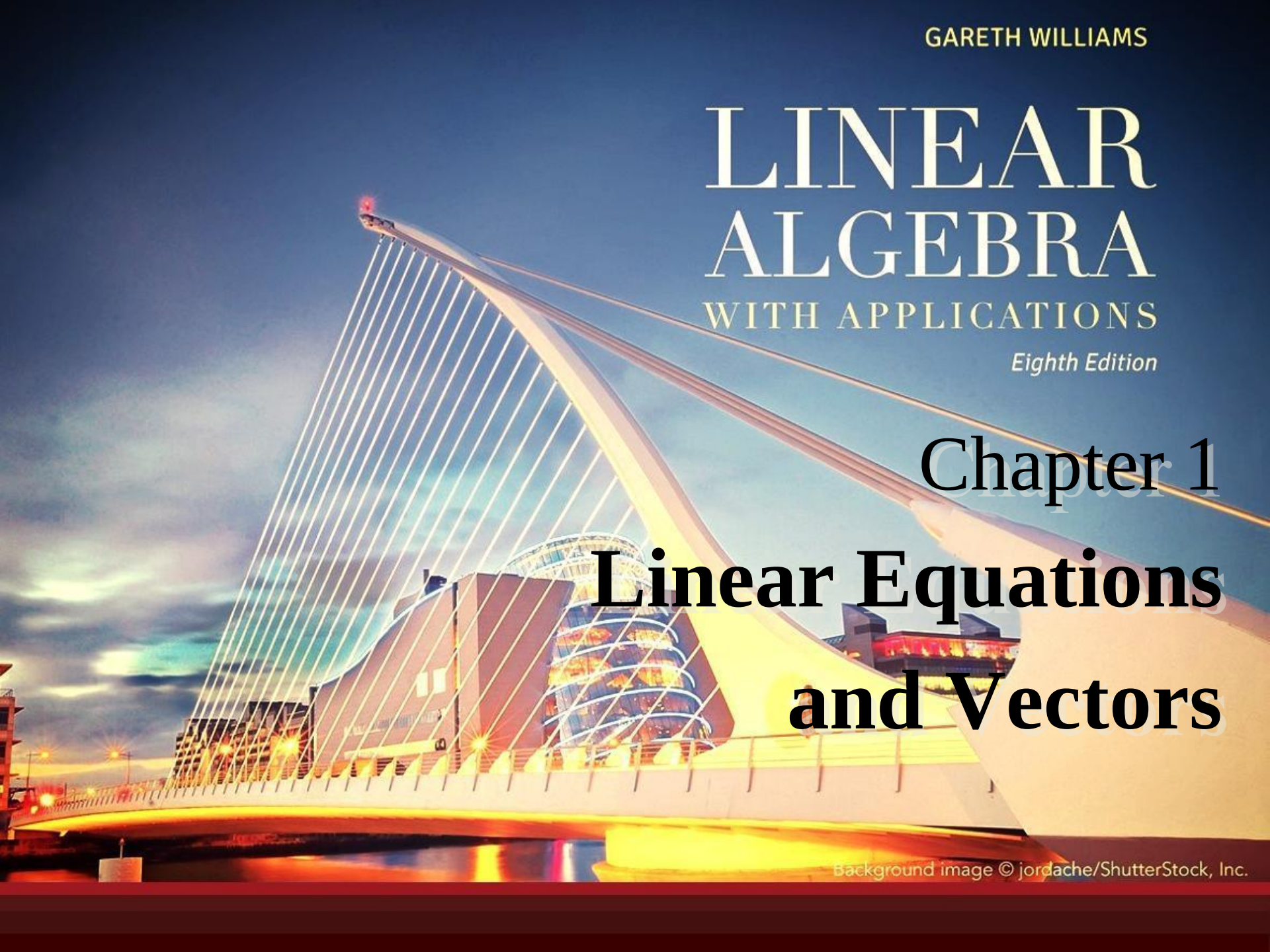




GARETH WILLIAMS

LINEAR ALGEBRA

WITH APPLICATIONS

Eighth Edition

Chapter 1

Linear Equations and Vectors

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1.The Importance of Mathematics and Its Application Tools:

Mathematics is a fundamental tool in many fields, and linear algebra plays an important role in modern mathematics.

2.Applications of Linear Algebra in Computer Science:

Linear algebra is applied in computer science in areas such as image processing, machine learning, data analysis, and cryptography.

3.Learning Content:

Studying mathematical theory, thinking logically with mathematics, and mastering the art of applying mathematics.

4.Systems of Linear Equations:

Linear algebra was developed to solve systems of linear equations, and its importance lies in understanding the behavior of solutions and their practical applications.

5.Vector Spaces:

The solutions to systems of linear equations form subsets of vector spaces, which are studied as fundamental algebraic structures.

6.Practical Application Cases:

Curve analysis and traffic flow analysis in road networks.

1.1 Matrices and Systems of Linear Equations

- Unique solution

$$x + y = 5$$

$$2x - y = 4$$

Write as $y = -x + 5$ and $y = 2x - 4$. The lines have slopes -1 and 2 , and y -intercepts 5 and -4 . They intersect at a point, the solution. There is a unique solution, $x = 3$, $y = 2$.

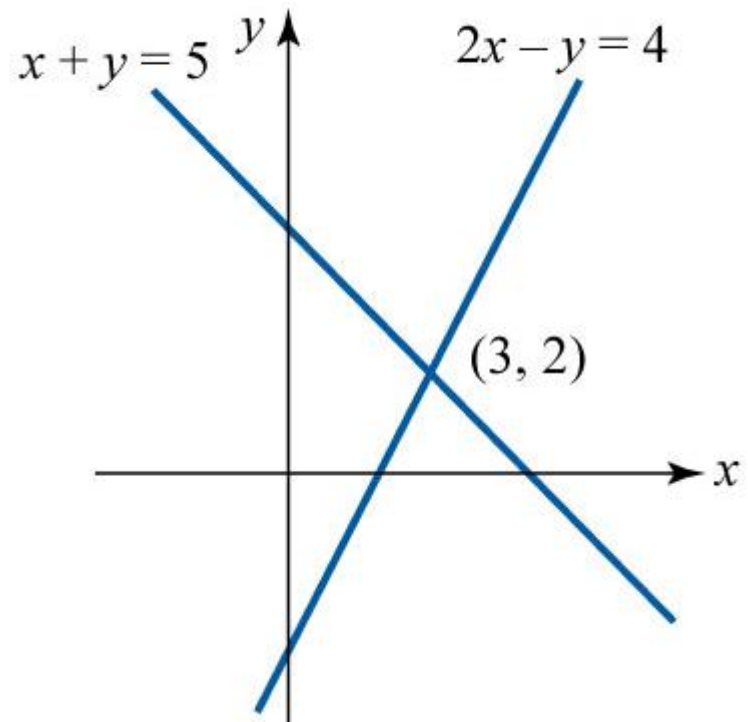


Figure 1.1

- No solution

$$-2x + y = 3$$

$$-4x + 2y = 4$$

Write as $y = 2x + 3$ and $y = 2x + 1$. The lines have slope 2, and y-intercepts 3 and 1. They are parallel. There is no point of intersection. No solution.

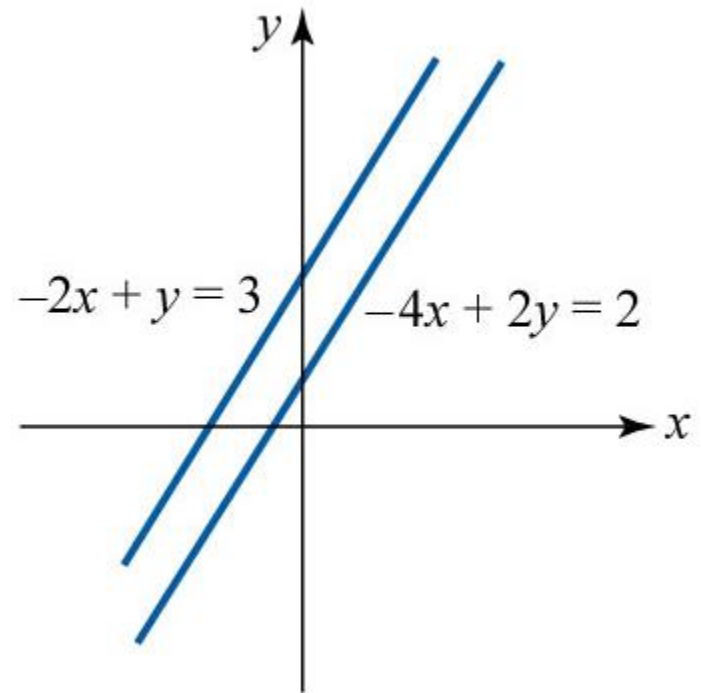


Figure 1.2

- Many solution

$$4x - 2y = 6$$

$$6x - 3y = 9$$

Each equation can be written as $y = 2x - 3$. The graph of each equation is a line with slope 2 and y-intercept -3 . Any point on the line is a solution. Many solutions.

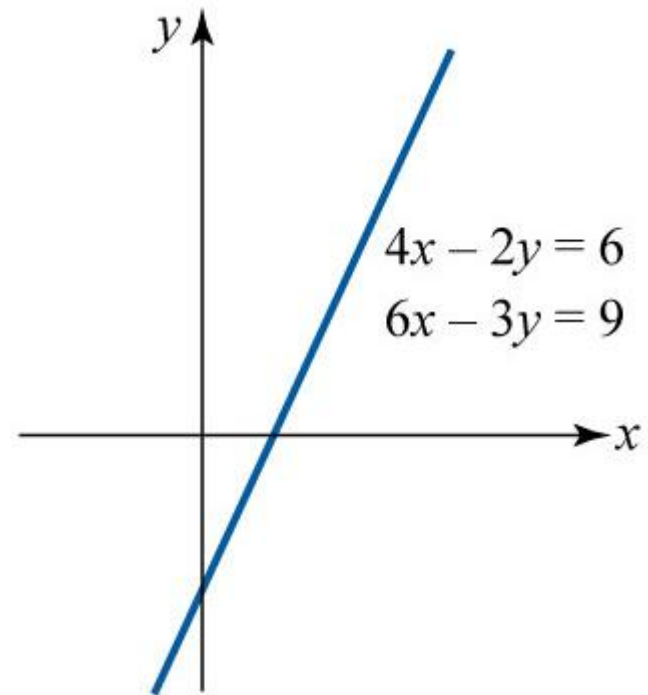
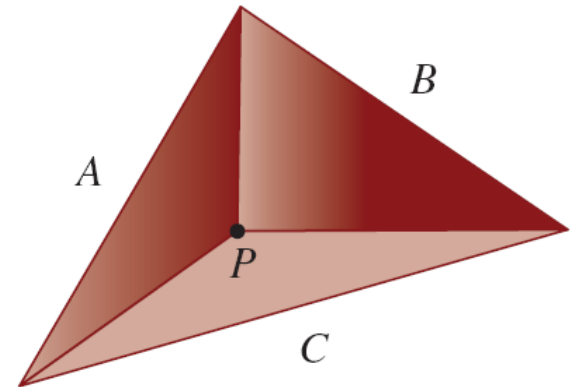
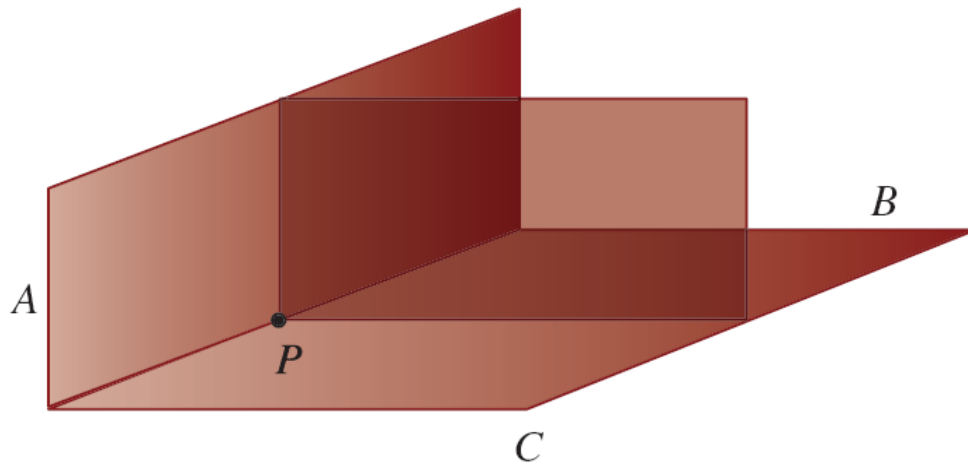


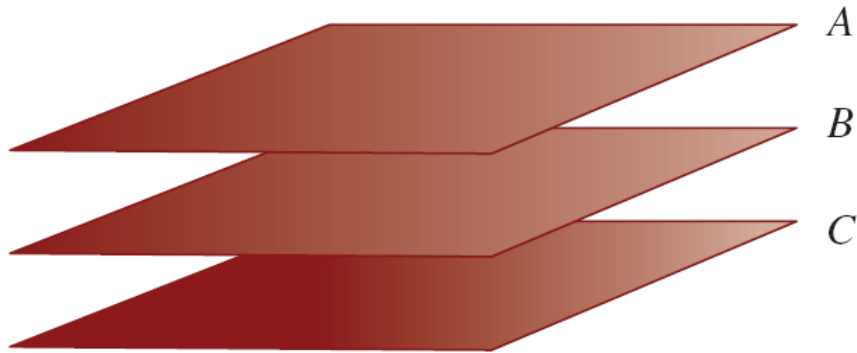
Figure 1.3

⊕ *Unique solution*

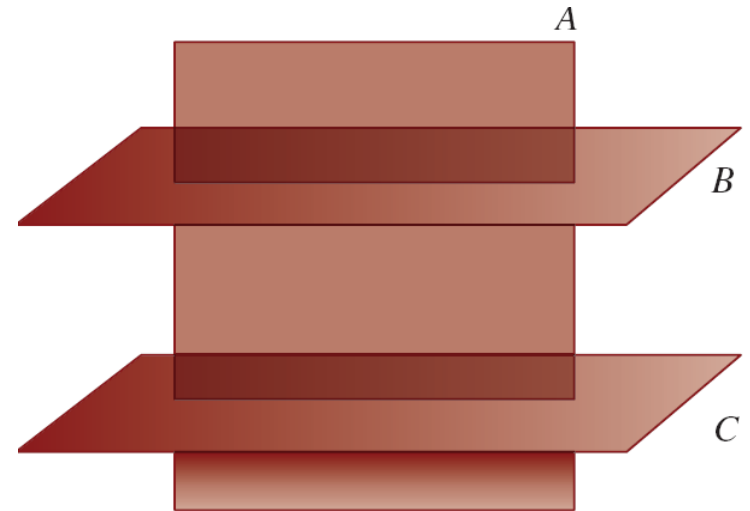


Three planes A , B , and C intersect at a single point P .
 P corresponds to a unique solution.

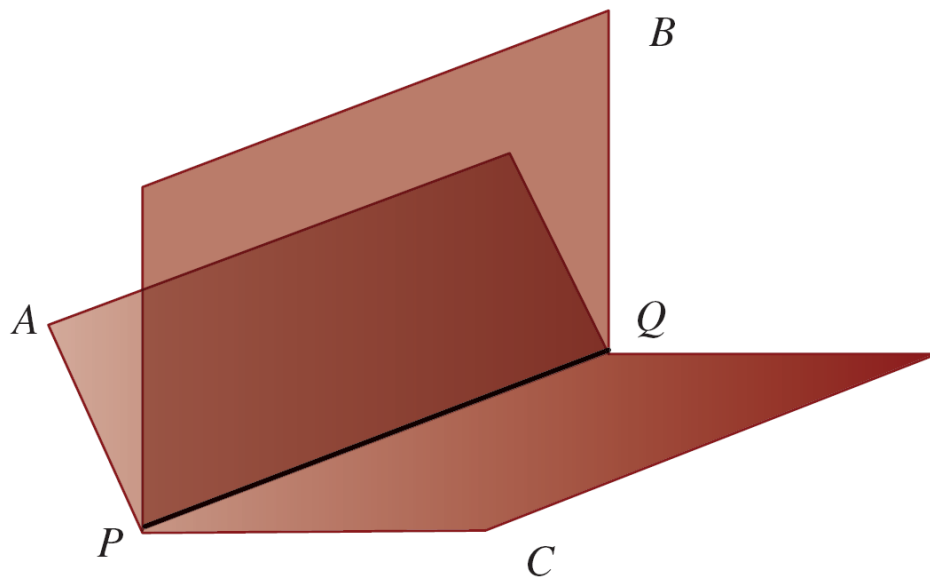
⊕ *No solutions*



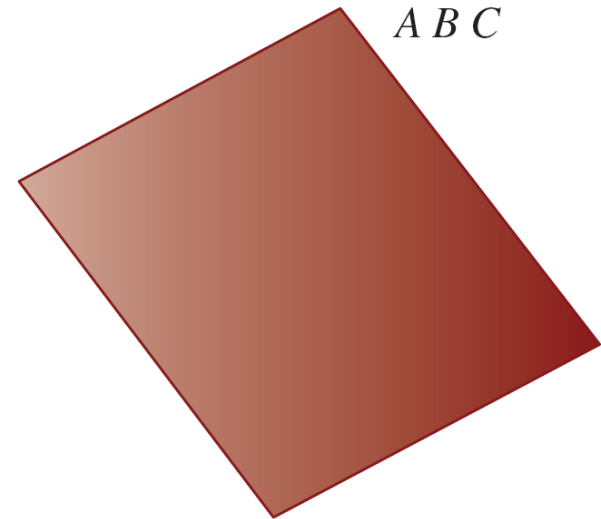
Planes A , B , and C have no points in common.
There is no solution.



⊕ *Many solutions*



Three planes A , B , and C intersect in a line PQ . Any point on the line is a solution.



Three equations represent the same plane. Any point on the plane is a solution.

Definition

A **matrix** is a rectangular array of numbers. The numbers in the array are called the *elements* of the matrix.

⊕ Rows and Columns

$$\begin{bmatrix} 2 & 3 & -4 \\ 7 & 5 & -1 \end{bmatrix}$$

$$[2 \quad 3 \quad -4]$$

row 1

$$[7 \quad 5 \quad -1]$$

row 2

$$\begin{bmatrix} 2 \\ 7 \end{bmatrix}$$

column 1

$$\begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

column 2

$$\begin{bmatrix} -4 \\ -1 \end{bmatrix}$$

column 3

⊕ Submatrix

$$A = \begin{bmatrix} 1 & 7 & 4 \\ 2 & 3 & 0 \\ 5 & 1 & -2 \end{bmatrix}$$

matrix A

$$P = \begin{bmatrix} 1 & 7 \\ 2 & 3 \\ 5 & 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

$$R = \begin{bmatrix} 1 & 4 \\ 5 & -2 \end{bmatrix}$$

submatrices of A

⊕ Size and Type

$$\begin{bmatrix} 1 & 0 & 3 \\ -2 & 4 & 5 \end{bmatrix}$$

2×3 matrix

$$\begin{bmatrix} 2 & 5 & 7 \\ -9 & 0 & 1 \\ -3 & 5 & 8 \end{bmatrix}$$

3×3 matrix

a square matrix

$$[4 \quad -3 \quad 8 \quad 5]$$

1×4 matrix

a row matrix

$$\begin{bmatrix} 8 \\ 3 \\ 2 \end{bmatrix}$$

3×1 matrix

a column matrix

⊕ Location

$\begin{bmatrix} 2 & 3 & -4 \\ 7 & 5 & -1 \end{bmatrix}$ The element 7 is in row 2, column 1.
We say that it is in location (2, 1).

⊕ Identity Matrices

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

⊕ matrix of coefficient and augmented matrix

$$x_1 + x_2 + x_3 = 2$$

$$2x_1 + 3x_2 + x_3 = 3$$

$$x_1 - x_2 - 2x_3 = -6$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & -1 & -2 \end{bmatrix}$$

matrix of coefficient

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 \\ 1 & -1 & -2 & -6 \end{bmatrix}$$

augmented matrix

Elementary Transformations

1. Interchange two equations.
2. Multiply both sides of an equation by a nonzero constant.
3. Add a multiple of one equation to another equation.

Elementary Row Operations

1. Interchange two rows of a matrix.
2. Multiply the elements of a row by a nonzero constant.
3. Add a multiple of the elements of one row to the corresponding elements of another row.

Example 1

Solving the following system of linear equation.

$$\begin{aligned}x_1 + x_2 + x_3 &= 2 \\2x_1 + 3x_2 + x_3 &= 3 \\x_1 - x_2 - 2x_3 &= -6\end{aligned}$$

Solution

Equation Method

Initial system:

$$\begin{array}{l} \text{Eq2} + (-2)\text{Eq1} \rightarrow 2x_1 + 3x_2 + x_3 = 3 \\ \text{Eq3} + (-1)\text{Eq1} \rightarrow x_1 - x_2 - 2x_3 = -6 \end{array}$$



$$\begin{array}{l} \text{Eq1} + (-1)\text{Eq2} \rightarrow x_1 + x_2 + x_3 = 2 \\ \text{Eq3} + (2)\text{Eq2} \rightarrow -2x_2 - 3x_3 = -8 \end{array}$$

Analogous Matrix Method

Augmented matrix:

$$\begin{array}{l} \text{R2} + (-2)\text{R1} \\ \text{R3} + (-1)\text{R1} \end{array} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 \\ 1 & -1 & -2 & -6 \end{bmatrix}$$



$$\begin{array}{l} \text{Eq1} + (-1)\text{Eq2} \\ \text{Eq3} + (2)\text{Eq2} \end{array} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & -2 & -3 & -8 \end{bmatrix}$$



$$x_1 + 2x_3 = 3$$

$$x_2 - x_3 = -1$$

$$(-1/5)\text{Eq3} \longrightarrow -5x_3 = -10$$



$$\begin{array}{l} \text{Eq1} + (-2)\text{Eq3} \\ \text{Eq2} + \text{Eq3} \end{array} \longrightarrow \begin{array}{l} x_1 + 2x_3 = 3 \\ x_2 - x_3 = -1 \\ x_3 = 2 \end{array}$$



$$x_1 = -1$$

$$x_2 = 1$$

$$x_3 = 2$$

The solution is

$$x_1 = -1, x_2 = 1, x_3 = 2.$$



$$(-1/5)\text{R3} \longrightarrow \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -5 & -10 \end{bmatrix}$$



$$\begin{array}{l} \text{R1} + (-2)\text{R3} \\ \text{R2} + \text{R3} \end{array} \longrightarrow \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

The solution is

$$x_1 = -1, x_2 = 1, x_3 = 2.$$

Example 2

Solving the following system of linear equation.

$$\begin{aligned}x_1 - 2x_2 + 4x_3 &= 12 \\2x_1 - x_2 + 5x_3 &= 18 \\-x_1 + 3x_2 - 3x_3 &= -8\end{aligned}$$

Solution

$$\begin{bmatrix} 1 & -2 & 4 & 12 \\ 2 & -1 & 5 & 18 \\ -1 & 3 & -3 & -8 \end{bmatrix} \begin{array}{l} \xrightarrow{\text{R2}+(-2)\text{R1}} \\ \xrightarrow{\text{R3}+\text{R1}} \end{array} \begin{bmatrix} 1 & -2 & 4 & 12 \\ 0 & 3 & -3 & -6 \\ 0 & 1 & 1 & 4 \end{bmatrix} \xleftarrow{\left(\frac{1}{3}\right)\text{R2}}$$

$$\begin{bmatrix} 1 & -2 & 4 & 12 \\ 0 & 1 & -1 & -2 \\ 0 & 1 & 1 & 4 \end{bmatrix} \begin{array}{l} \xrightarrow{\text{R1}+(2)\text{R2}} \\ \xrightarrow{\text{R3}+(-1)\text{R2}} \end{array} \begin{bmatrix} 1 & 0 & 2 & 8 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 2 & 6 \end{bmatrix} \xleftarrow{\left(\frac{1}{2}\right)\text{R3}}$$

$$\begin{bmatrix} 1 & 0 & 2 & 8 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{array}{l} \xrightarrow{\text{R1}+(-2)\text{R3}} \\ \xrightarrow{\text{R2}+\text{R3}} \end{array} \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{bmatrix} \quad \begin{array}{l} \text{The solution is} \\ x_1 = 2, x_2 = 1, x_3 = 3 \end{array}$$

Example 3

Solving the system

$$4x_1 + 8x_2 - 12x_3 = 44$$

$$3x_1 + 6x_2 - 8x_3 = 32$$

$$-2x_1 - 3x_2 = -7$$

Solution

$$\begin{bmatrix} 4 & 8 & -12 & 44 \\ 3 & 6 & -8 & 32 \\ -2 & -1 & 0 & -7 \end{bmatrix} \xrightarrow{(1/4)R1} \begin{bmatrix} 1 & 2 & -3 & 11 \\ 3 & 6 & -8 & 32 \\ -2 & -1 & 0 & -7 \end{bmatrix}$$

$$\xrightarrow[\text{R3}+2\text{R1}]{\text{R2}+(-3)\text{R1}} \begin{bmatrix} 1 & 2 & -3 & 11 \\ 0 & 0 & 1 & -1 \\ 0 & 3 & -6 & 15 \end{bmatrix}$$

$$\begin{array}{ccc}
\begin{bmatrix} 1 & 2 & -3 & 11 \\ 0 & 0 & 1 & -1 \\ 0 & 3 & -6 & 15 \end{bmatrix} & \xrightarrow{\text{R2} \leftrightarrow \text{R3}} & \begin{bmatrix} 1 & 2 & -3 & 11 \\ 0 & 3 & -6 & 15 \\ 0 & 0 & 1 & -1 \end{bmatrix} \\
\\
& \xrightarrow{(1/3)\text{R2}} & \begin{bmatrix} 1 & 2 & -3 & 11 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 1 & -1 \end{bmatrix} \\
\\
& \xrightarrow{\text{R1} + (-2)\text{R2}} & \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 1 & -1 \end{bmatrix} \\
\\
& \xrightarrow[\text{R2} + 2\text{R3}]{\text{R1} + (-1)\text{R3}} & \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \end{bmatrix}.
\end{array}$$

The solution is $x_1 = 2$, $x_2 = 3$, $x_3 = -1$.

Example 4

Solving the following three systems of linear equation, all of which have the same matrix of coefficients.

$$\begin{aligned}x_1 - x_2 + 3x_3 &= b_1 \\ 2x_1 - x_2 + 4x_3 &= b_2 \\ -x_1 + 2x_2 - 4x_3 &= b_3\end{aligned} \quad \text{for } \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 11 \\ -11 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix} \text{ in turn}$$

Solution

$$\begin{bmatrix} 1 & -1 & 3 & 8 & 0 & 3 \\ 2 & -1 & 4 & 11 & 1 & 3 \\ -1 & 2 & -4 & -11 & 2 & -4 \end{bmatrix} \begin{array}{l} \xrightarrow{\text{R2}+(-2)\text{R1}} \\ \xrightarrow{\text{R3}+\text{R1}} \end{array}$$

$$\begin{bmatrix} 1 & -1 & 3 & 8 & 0 & 3 \\ 0 & 1 & -2 & -5 & 1 & -3 \\ 0 & 1 & -1 & -3 & 2 & -1 \end{bmatrix} \xrightarrow[\text{R3}+(-1)\text{R2}]{\text{R1}+\text{R2}} \begin{bmatrix} 1 & 0 & 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & -5 & 1 & -3 \\ 0 & 0 & 1 & 2 & 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & -5 & 1 & -3 \\ 0 & 0 & 1 & 2 & 1 & 2 \end{bmatrix} \xrightarrow[\text{R2}+2\text{R3}]{\text{R1}+(-1)\text{R3}} \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & -2 \\ 0 & 1 & 0 & -1 & 3 & 1 \\ 0 & 0 & 1 & 2 & 1 & 2 \end{bmatrix}$$

The solutions are $x_1 = 1, x_2 = -1, x_3 = 2$
 $x_1 = 0, x_2 = 3, x_3 = 1$
 $x_1 = -2, x_2 = 1, x_3 = 2.$

1.2 Gauss-Jordan Elimination

Definition

A matrix is in **reduced echelon form** if

1. Any rows consisting entirely of zeros are grouped at the bottom of the matrix.
2. The first nonzero element of each other row is 1. This element is called a **leading 1**.
3. The leading 1 of each after the first is positioned to the right of the leading 1 of the previous row.
4. All other elements in a column that contains a leading 1 are zero.

In Reduced Echelon Form

$$\begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 5 & 0 & 0 & 8 \\ 0 & 1 & 7 & 0 & 0 & 9 \\ 0 & 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & 3 & 0 & 4 \\ 0 & 0 & 1 & 2 & 0 & 7 \\ 0 & 0 & 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Not in Reduced Echelon Form

$$\begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Row of zeros
not at bottom
of matrix

$$\begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 3 & 4 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

First nonzero
element in row
2 is not 1

$$\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 1 & 0 & 3 \end{bmatrix}$$

Leading 1 in
row 3 not to the
right of leading
1 in row 2

$$\begin{bmatrix} 1 & 7 & 0 & 8 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Nonzero
element above
leading 1 in
Row 2

Example 1

Use the method of Gauss-Jordan elimination to find reduced echelon form of the following matrix.

Solution

$$\begin{bmatrix} 0 & 0 & 2 & -2 & 2 \\ 3 & 3 & -3 & 9 & 12 \\ 4 & 4 & -2 & 11 & 12 \end{bmatrix}$$

$$\begin{array}{l} \approx \\ R1 \leftrightarrow R2 \end{array} \begin{array}{c} \text{pivot} \\ \swarrow \\ \textcircled{3} \end{array} \begin{bmatrix} 3 & 3 & -3 & 9 & 12 \\ 0 & 0 & 2 & -2 & 2 \\ 4 & 4 & -2 & 11 & 12 \end{bmatrix} \xrightarrow{(1/3)R1} \begin{bmatrix} 1 & 1 & -1 & 3 & 4 \\ 0 & 0 & 2 & -2 & 2 \\ 4 & 4 & -2 & 11 & 12 \end{bmatrix}$$

$$\xrightarrow{R3 + (-4)R1} \begin{bmatrix} 1 & 1 & -1 & 3 & 4 \\ 0 & 0 & \textcircled{2} & -2 & 2 \\ 0 & 0 & 2 & -1 & -4 \end{bmatrix} \xrightarrow{(1/2)R2} \begin{bmatrix} 1 & 1 & -1 & 3 & 4 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 2 & -1 & -4 \end{bmatrix}$$

pivot

$$\begin{bmatrix} 1 & 1 & -1 & 3 & 4 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 2 & -1 & -4 \end{bmatrix} \xrightarrow[\text{R3}+(-2)\text{R2}]{\text{R1}+\text{R2}} \begin{bmatrix} 1 & 1 & 0 & 2 & 5 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & \textcircled{1} & -6 \end{bmatrix}$$


 pivot

$$\xrightarrow[\text{R2}+\text{R3}]{\text{R1}+(-2)\text{R3}} \begin{bmatrix} 1 & 1 & 0 & 0 & 17 \\ 0 & 0 & 1 & 0 & -5 \\ 0 & 0 & 0 & 1 & -6 \end{bmatrix}$$

The matrix is the reduced echelon form of the given matrix.

Example 2

Solve, if possible, the system of equations

$$3x_1 - 3x_2 + 3x_3 = 9$$

$$2x_1 - x_2 + 4x_3 = 7$$

$$3x_1 - 5x_2 - x_3 = 7$$

Solution

$$\begin{bmatrix} \textcircled{3} & -3 & 3 & 9 \\ 2 & -1 & 4 & 7 \\ 3 & -5 & -1 & 7 \end{bmatrix} \xrightarrow{(1/3)R1} \begin{bmatrix} \textcircled{1} & -1 & 1 & 3 \\ 2 & -1 & 4 & 7 \\ 3 & -5 & -1 & 7 \end{bmatrix}$$

$$\xrightarrow[\text{R3}+(-3)\text{R1}]{\text{R2}+(-2)\text{R1}} \begin{bmatrix} 1 & -1 & 1 & 3 \\ 0 & \textcircled{1} & 2 & 1 \\ 0 & -2 & -4 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 1 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & -2 & -4 & -2 \end{bmatrix} \xrightarrow[\text{R3} + (2)\text{R2}]{\text{R1} + \text{R2}} \begin{bmatrix} 1 & 0 & 3 & 4 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{array}{l} x_1 + 3x_3 = 4 \\ x_2 + 2x_3 = 1 \end{array} \Rightarrow \begin{array}{l} x_1 = -3x_3 + 4 \\ x_2 = -2x_3 + 1 \end{array}$$

The general solution to the system is

$$x_1 = -3r + 4$$

$$x_2 = -2r + 1$$

$$x_3 = r$$

which r is real number (called a parameter).

Example 3

This example illustrates that the general solution can involve a number of parameters. Solve the system of equations

$$x_1 + 2x_2 - x_3 + 3x_4 = 4$$

$$2x_1 + 4x_2 - 2x_3 + 7x_4 = 10$$

$$-x_1 - 2x_2 + x_3 - 4x_4 = -6$$

Solution

$$\begin{bmatrix} 1 & 2 & -1 & 3 & 4 \\ 2 & 4 & -2 & 7 & 10 \\ -1 & -2 & 1 & -4 & -6 \end{bmatrix} \xrightarrow[\text{R3+R1}]{\text{R2+}(-2)\text{R1}} \begin{bmatrix} 1 & 2 & -1 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -1 & -2 \end{bmatrix}$$

$$\xrightarrow[\text{R3+R2}]{\text{R1+}(-3)\text{R2}} \begin{bmatrix} 1 & 2 & -1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

We have arrived at the reduced echelon form. The corresponding system of equations is

$$x_1 + 2x_2 - x_3 = -2$$

$$x_4 = 2$$

Expressing the leading variables in terms of the remaining variables we get

$$x_1 = -2x_2 + x_3 - 2, \quad x_4 = 2$$

Let us assign the arbitrary values r to x_2 and s to x_3 . The general solution is

$$x_1 = -2r + s - 2, \quad x_2 = r, \quad x_3 = s, \quad x_4 = 2$$

Specific solutions can be obtained by giving r and s various values.

Example 4

This example illustrates a system that has no solution. Let us try to solve the system

$$x_1 + x_2 + 5x_3 = 3$$

$$x_2 + 3x_3 = -1$$

$$x_1 + 2x_2 + 8x_3 = 3$$

Solution

$$\begin{bmatrix} 1 & 1 & 5 & 3 \\ 0 & 1 & 3 & -1 \\ 1 & 2 & 8 & 3 \end{bmatrix} \xrightarrow{R3+(-1)R1} \begin{bmatrix} 1 & 1 & 5 & 3 \\ 0 & 1 & 3 & -1 \\ 0 & 1 & 3 & 0 \end{bmatrix}$$

$$\xrightarrow[\begin{matrix} R1+(-1)R2 \\ R3+(-1)R2 \end{matrix}]{\begin{bmatrix} 1 & 0 & 2 & 4 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}} \xrightarrow[\begin{matrix} R1+(-4)R3 \\ R2+R3 \end{matrix}]{\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}$$

The last row of this reduced echelon form gives the equation

$$0x_1 + 0x_2 + 0x_3 = 1$$

This equation cannot be satisfied for any values of x_1, x_2 and x_3 . Thus the system has no solution. (This information was in fact available from the next-to-last matrix.)

Homogeneous System of linear Equations

Theorem 1.1

A homogeneous system of linear equations in n variables always has the solution $x_1 = 0, x_2 = 0, \dots, x_n = 0$. This solution is called the **trivial solution**.

Theorem 1.2

A homogeneous system of linear equations that has more variables than equations has many solutions. One of these solutions is the trivial solution.

1.3 The Vector Space R^n

Rectangular Coordinate System

- the **origin** : $(0, 0)$
- the **position vector** : $\vec{OA}$
- the **initial point** : O
- the **terminal point** : $A(5, 3)$

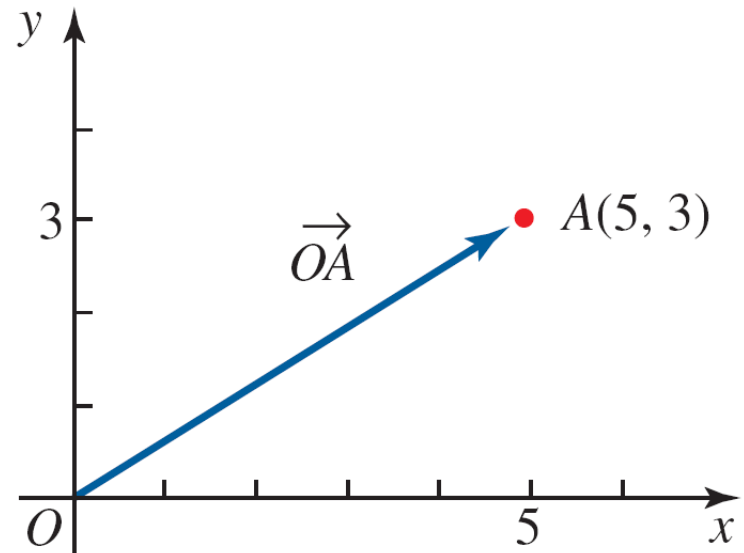


Figure 1.5

EXAMPLE 1

Sketch the position vectors $\overrightarrow{OA} = (4, 1)$, $\overrightarrow{OB} = (-5, -2)$ and $\overrightarrow{OC} = (-3, 4)$. See Figure 1.6.

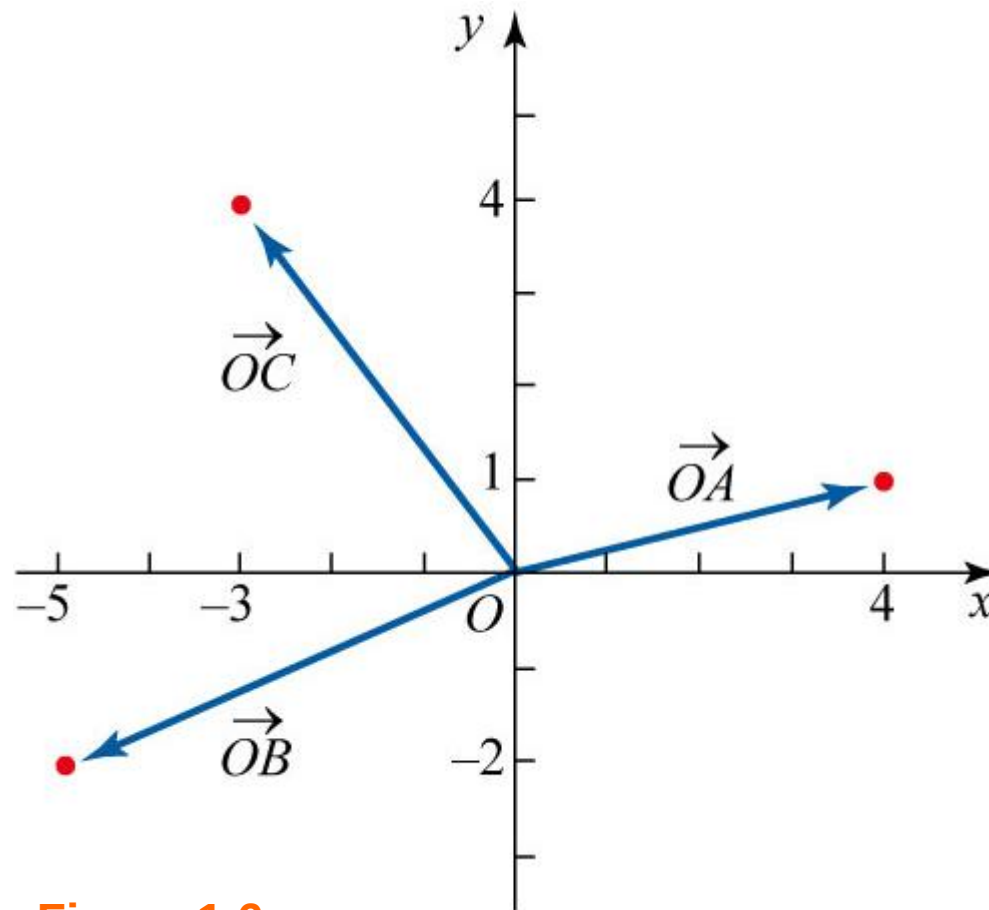


Figure 1.6

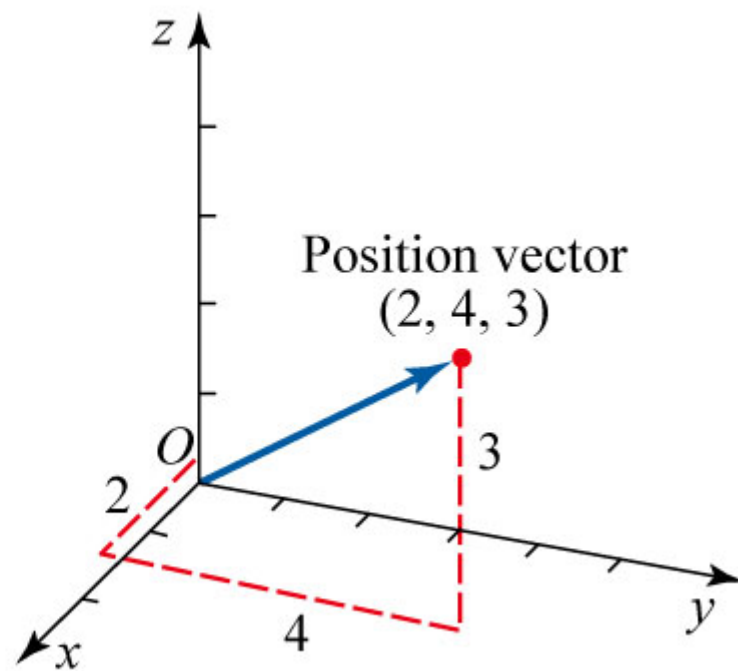
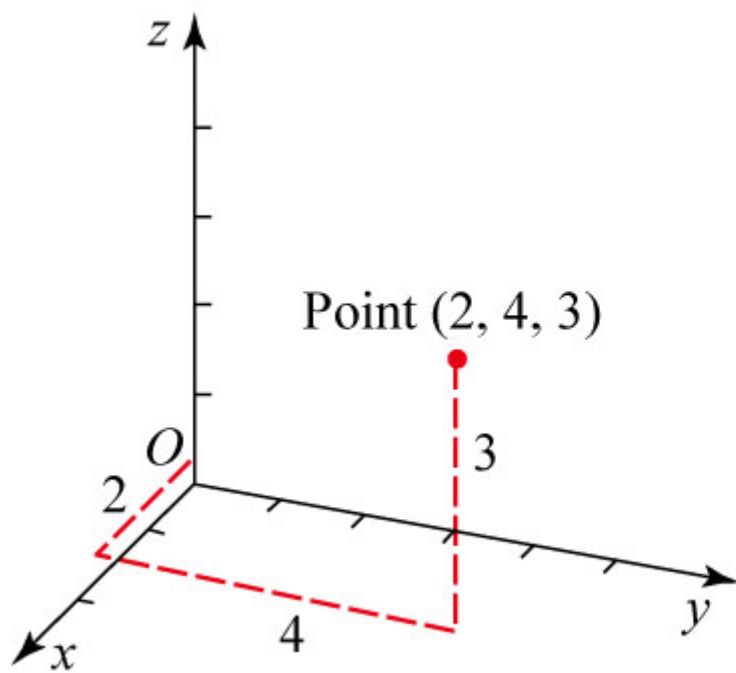


Figure 1.7

Let $(u_1, u_2, \dots, u_n)$ be a sequence of n real numbers. The set of all such sequences is called n -space and is called $\mathbf{R}^n$. u_1 is the **first component** of $(u_1, u_2, \dots, u_n)$, u_2 is the **second component** and so on.

Example

$\mathbf{R}^4$ is the collection of all sets of four ordered real numbers. For example, $(1, 2, 3, 4)$ and $(-1, 3, 5.2, 0)$ are element of $\mathbf{R}^4$.

$\mathbf{R}^5$ is the collection of all sets of five ordered real numbers. For example, $(-1, 2, 0, 3, 9)$ is in this collection.

Addition and Scalar multiplication

Let $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$ be two elements of $\mathbf{R}^n$. We say that $\mathbf{u}$ and $\mathbf{v}$ are **equal** if $u_1 = v_1, \dots, u_n = v_n$ (their **corresponding components** are equal.)

Definition

Let $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$ be elements of $\mathbf{R}^n$ and let c be a scalar. Addition and Scalar multiplication are performed as follows.

Addition: $\mathbf{u} + \mathbf{v} = (u_1 + v_1, \dots, u_n + v_n)$

Scalar multiplication : $c\mathbf{u} = (cu_1, \dots, cu_n)$

Example 2

Let $\mathbf{u} = (-1, 4, 3, 7)$ and $\mathbf{v} = (-2, -3, 1, 0)$ be vectors in $\mathbf{R}^4$.

Find $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u}$.

Solution

$$\mathbf{u} + \mathbf{v} = (-1, 4, 3, 7) + (-2, -3, 1, 0) = (-3, 1, 4, 7)$$

$$3\mathbf{u} = 3(-1, 4, 3, 7) = (-3, 12, 9, 21)$$

Example 3

This example gives us a geometrical interpretation of vector addition. Consider the sum of the vectors $(4, 1)$ and $(2, 3)$. We get

$$(4, 1) + (2, 3) = (6, 4)$$

In Figure 1.8 we interpret these vectors as position vectors. Construct the parallelogram having the vectors $(4, 1)$ and $(2, 3)$ as adjacent sides. The vector $(6, 4)$, the sum, will be the diagonal of the parallelogram.

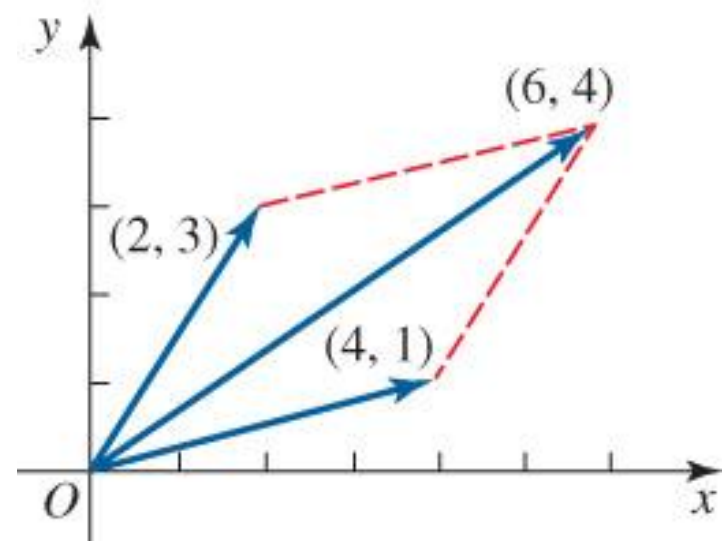


Figure 1.8

Such vectors are used in the physical sciences to describe forces. In this example $(4, 1)$ and $(2, 3)$ might be forces, acting on a body at origin, O . The vectors would give the directions of the forces and their lengths would be the magnitudes of the forces, $\sqrt{4^2 + 1^2} = 4.12$ and $\sqrt{2^2 + 3^2} = 3.61$. The vector sum $(6, 4)$ is the resultant force, a single force that would be equivalent to the two forces. The magnitude of the resultant force would be $\sqrt{6^2 + 4^2} = 7.21$.

In general, if $\mathbf{u}$ and $\mathbf{v}$ are vectors in the same vector space, then $\mathbf{u} + \mathbf{v}$ is the diagonal of the parallelogram defined by $\mathbf{u}$ and $\mathbf{v}$. See Figure 1.9.

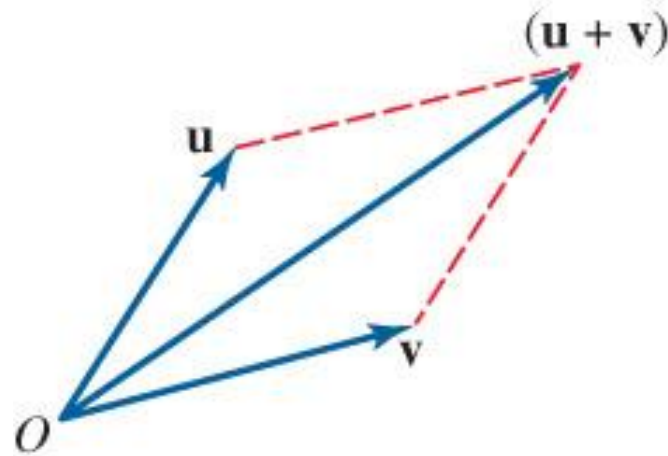


Figure 1.9

Example 4

This example gives us a geometrical interpretation of scalar multiplication. Consider the scalar multiple of the vector $(3, 2)$ by 2. We get

$$2(3, 2) = (6, 4)$$

Observe in Figure 1.10 that $(6, 4)$ is a vector in the same direction as $(3, 2)$, and 2 times it in the length.

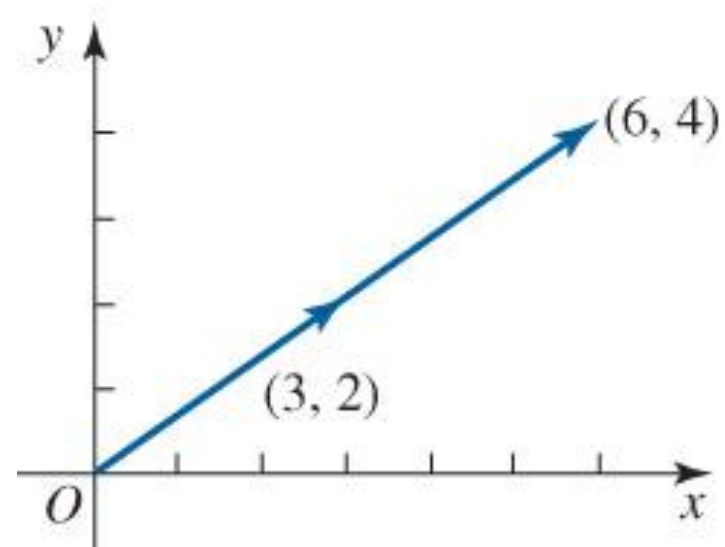
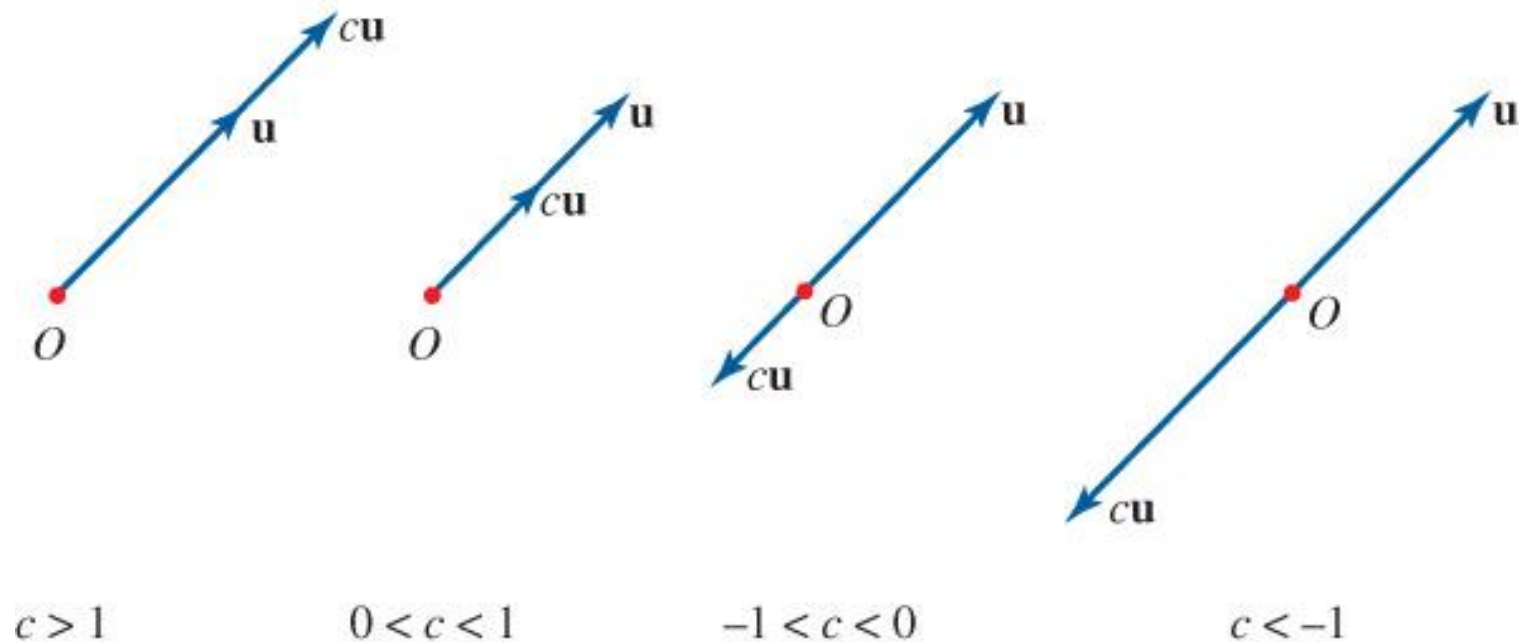


Figure 1.10

In general, if $\mathbf{u}$ is a vector in any vector space, and c a nonzero scalar, the direction of $c\mathbf{u}$ will be the same as the direction of $\mathbf{u}$ if $c > 0$, and the opposite direction to $\mathbf{u}$ if $c < 0$. The length of $c\mathbf{u}$ is $|c|$ times the length of $\mathbf{u}$. See Figure 1.11

Figure 1.11



Special Vectors

The vector $(0, 0, \dots, 0)$, having n zero components, is called the **zero vector** of $\mathbf{R}^n$ and is denoted $\mathbf{0}$.

Negative Vector

The vector $(-1)\mathbf{u}$ is written $-\mathbf{u}$ and is called the **negative** of $\mathbf{u}$.

Subtraction

Subtraction is performed on elements of $\mathbf{R}^n$ by subtracting corresponding components. For example, in $\mathbf{R}^3$

$$(5, 3, -6) - (2, 1, 3) = (3, 2, -9)$$

Theorem 1.3

Properties of Vector Addition and Scalar Multiplication

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in $\mathbf{R}^n$ and let c and d be scalars.

(a) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

Commutative property

(b) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$

Associative property

(c) $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$

Property of the zero vector

(d) $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$

Property of the negative vector

(e) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$

(f) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$

(g) $c(d\mathbf{u}) = (cd)\mathbf{u}$

} Distributive properties

(h) $1\mathbf{u} = \mathbf{u}$

Scalar multiplication by 1

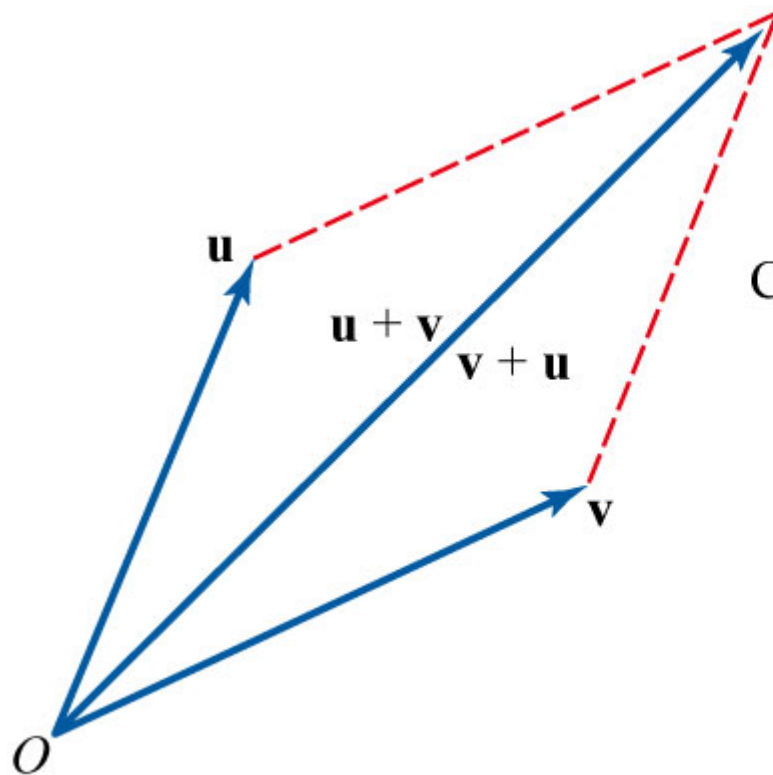
Proof These result are proved by writing the vectors in terms of components and using the definitions of vector addition and scalar multiplication, and the properties of real numbers. We give the proofs of (a) and (e).

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u} :$$

Let $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$.Then

$$\begin{aligned}\mathbf{u} + \mathbf{v} &= (u_1, \dots, u_n) + (v_1, \dots, v_n) \\ &= (u_1 + v_1, \dots, u_n + v_n) \\ &= (v_1 + u_1, \dots, v_n + u_n) \\ &= (v_1, \dots, v_n) + (u_1, \dots, u_n) \\ &= \mathbf{v} + \mathbf{u}\end{aligned}$$

$$\begin{aligned}c(\mathbf{u} + \mathbf{v}) &= c((u_1, \dots, u_n) + (v_1, \dots, v_n)) \\ &= c(u_1 + v_1, \dots, u_n + v_n) \\ &= (c(u_1 + v_1), \dots, c(u_n + v_n)) \\ &= (cu_1 + cv_1, \dots, cu_n + cv_n) \\ &= (cu_1, \dots, cu_n) + (cv_1, \dots, cv_n) \\ &= c(u_1, \dots, u_n) + c(v_1, \dots, v_n) \\ &= c\mathbf{u} + c\mathbf{v}\end{aligned}$$



Commutativity of vector addition
 $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

Figure 1.12

Linear Combinations of Vectors

Example

Let $\mathbf{u} = (2, 5, -3)$, $\mathbf{v} = (-4, 1, 9)$, and $\mathbf{w} = (4, 0, 2)$. Determine the vector combination $2\mathbf{u} - 3\mathbf{v} + \mathbf{w}$.

Solution

$$\begin{aligned} 2\mathbf{u} - 3\mathbf{v} + \mathbf{w} &= 2(2, 5, -3) - 3(-4, 1, 9) + (4, 0, 2) \\ &= (4, 10, -6) - (-12, 3, 27) + (4, 0, 2) \\ &= (4 + 12 + 4, 10 - 3 + 0, -6 - 27 + 2) \\ &= (20, 7, -31) \end{aligned}$$

Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ be vectors in $\mathbf{R}^n$. The vector $\mathbf{v}$ in $\mathbf{R}^n$ is a *linear combination* of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ if there exist scalars $c_1, c_2, \dots, c_m$ such that $\mathbf{v}$ can be written

$$\mathbf{v} = c_1 \mathbf{v}_1 + \dots + c_m \mathbf{v}_m$$

Example 6

Determine whether the vector $(8, 0, 5)$ is a linear combination of the vectors $(1, 2, 3)$, $(0, 1, 4)$, and $(2, -1, 1)$.

Solution

$$c_1(1, 2, 3) + c_2(0, 1, 4) + c_3(2, -1, 1) = (8, 0, 5)$$

$$(c_1 + 2c_3, 2c_1 + c_2 - c_3, 3c_1 + 4c_2 + c_3) = (8, 0, 5)$$

$$c_1 + 2c_3 = 8$$

$$2c_1 + c_2 - c_3 = 0$$

$$3c_1 + 4c_2 + c_3 = 5$$

$$c_1 = 2, c_2 = -1, c_3 = 3$$

$$(8, 0, 5) = 2(1, 2, 3) - 1(0, 1, 4) + 3(2, -1, 1)$$

Column Vectors

We defined addition and scalar multiplication of column vectors in $\mathbf{R}^n$ in a componentwise manner:

$$\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix} \quad \text{and} \quad c \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} cu_1 \\ \vdots \\ cu_n \end{bmatrix}$$

For example, in $\mathbf{R}^2$,

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix} \quad \text{and} \quad 4 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}$$

1.4 Subspace of $\mathbf{R}^n$

The subset must be *closed under addition and under scalar multiplication*.

All the other vector space properties such as the commutative property $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$, and the associative property $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$, are inherited from larger $\mathbf{R}^n$.

A nonempty subset of the vector space $\mathbf{R}^n$ that has all the algebraic properties of $\mathbf{R}^n$ is called a *subspace*. A subset of $\mathbf{R}^n$ is a subspace if it is closed under addition and under scalar multiplication.

EXAMPLE 1

A subset of a vector space is often defined in terms of an arbitrary vector. Consider the subset W of $\mathbf{R}^3$ consisting of vectors of the form $(a, b, a+b)$, where the third component is the sum of the first two. The vector $(2, 5, 7)$ is in W for example, whereas $(2, 5, 9)$ is not. Let us show that W is a subspace of $\mathbf{R}^3$. Let $\mathbf{u} = (a, b, a+b)$ and $\mathbf{v} = (c, d, c+d)$ be vectors in W and k be a scalar. Then

$$\begin{aligned}\mathbf{u} + \mathbf{v} &= (a, b, a+b) + (c, d, c+d) \\ &= (a+c, b+d, a+b+c+d) \\ &= (a+c, b+d, (a+c) + (b+d))\end{aligned}$$

$\mathbf{u} + \mathbf{v}$ is in W since the third component is the sum of the first two components. W is closed under addition. Further,

$$\begin{aligned} k\mathbf{u} &= k(a, b, a + b) \\ &= (ka, kb, k(a + b)) \\ &= (ka, kb, ka + kb) \end{aligned}$$

$k\mathbf{u}$ is in W . W is also closed under scalar multiplication. W is a subspace of $\mathbf{R}^3$. It has all the algebraic properties of a vector space.

Let us look at the geometrical interpretation of this subspace W . Consider the arbitrary vector $\mathbf{u} = (a, b, a + b)$.

Separate the variable a and b as follows:

$$\begin{aligned}\mathbf{u} &= (a, b, a+b) \\ &= (a, 0, a) + (0, b, b) \\ &= a(1, 0, 1) + b(0, 1, 1)\end{aligned}$$

The vectors $(1, 0, 1)$ and $(0, 1, 1)$ define a plane through the origin. See Figure 1.13. Any linear combination of vectors $\mathbf{u} = a(1, 0, 1) + b(0, 1, 1)$ lies in this plane. W is in fact the plane determined by the vectors $(1, 0, 1)$ and $(0, 1, 1)$.

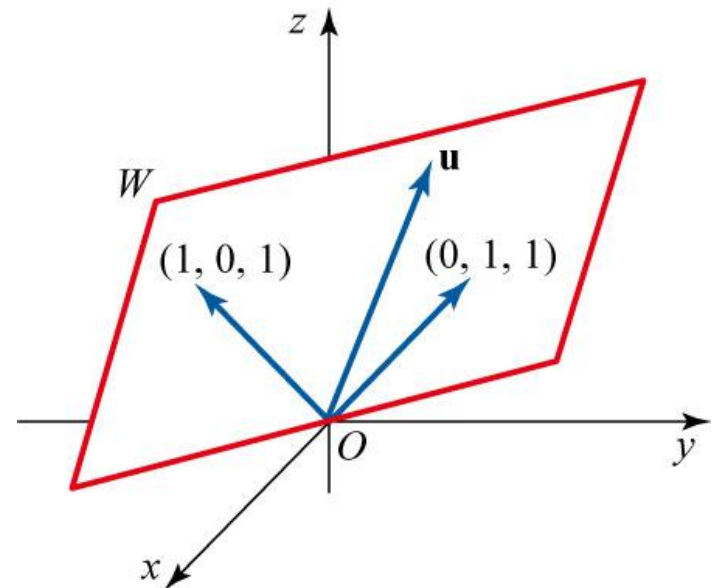


Figure 1.13 The plane determined by the vectors $(1, 0, 1)$ and $(0, 1, 1)$

When every vector of a vector space can be written as a linear combination of a set of vectors as here we say that vectors *span* the space. Thus the vectors $(1, 0, 1)$ and $(0, 1, 1)$ span W .

The preceding example leads to a general result. Let $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ be vectors in a plane W . The sum $\mathbf{u} + \mathbf{v}$ lies in the plane, and so does the scalar multiple $k\mathbf{w}$. See Figure 1.14. Thus W is a subspace.

A set of vectors that form a plane is a subspace.

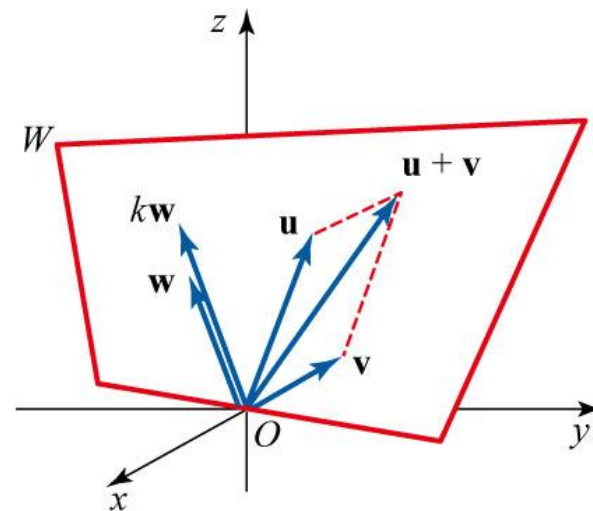


Figure 1.14 W is closed under addition and scalar multiplication.

EXAMPLE 2

Consider the subset V of $\mathbf{R}^3$ of vectors of the form $(a, 2a, 3a)$, where the second component is twice of the first, and the third is three times the first. Let us show that V is a subspace of $\mathbf{R}^3$. Let $(a, 2a, 3a)$ and $(b, 2b, 3b)$ be two vectors in V , and let k be a scalar. Then

$$\begin{aligned}(a, 2a, 3a) + (b, 2b, 3b) &= (a+b, 2a+2b, 3a+3b) \\ &= (a+b, 2(a+b), 3(a+b))\end{aligned}$$

This is a vector in V since the second component is twice the first, and the third is three times the first. V is closed under addition. Further,

$$k(a, 2a, 3a) = (ka, 2ka, 3ka)$$

The vector is in V . V is closed under scalar multiplication. V is a subspace of $\mathbf{R}^3$. We now proceed to look at the geometry. Consider the arbitrary vector $\mathbf{u}=(a, 2a, 3a)$. We get

$$\begin{aligned}\mathbf{u} &= (a, 2a, 3a) \\ &= a(1, 2, 3)\end{aligned}$$

The vectors of V are scalar multiples of the single vector $(1, 2, 3)$. These vectors form the line through the origin defined by the vector $(1, 2, 3)$. See Figure 1.15. The vector $(1, 2, 3)$ spans V .

The preceding example leads to a general result. Let $\mathbf{u}$ and $\mathbf{v}$ be vectors on a line V . The sum $\mathbf{u}+\mathbf{v}$ lies on the line, and so does the scalar multiple $k\mathbf{u}$. See Figure 1.16. Thus V is a subspace.

A set of vectors that forms a line is a subspace.

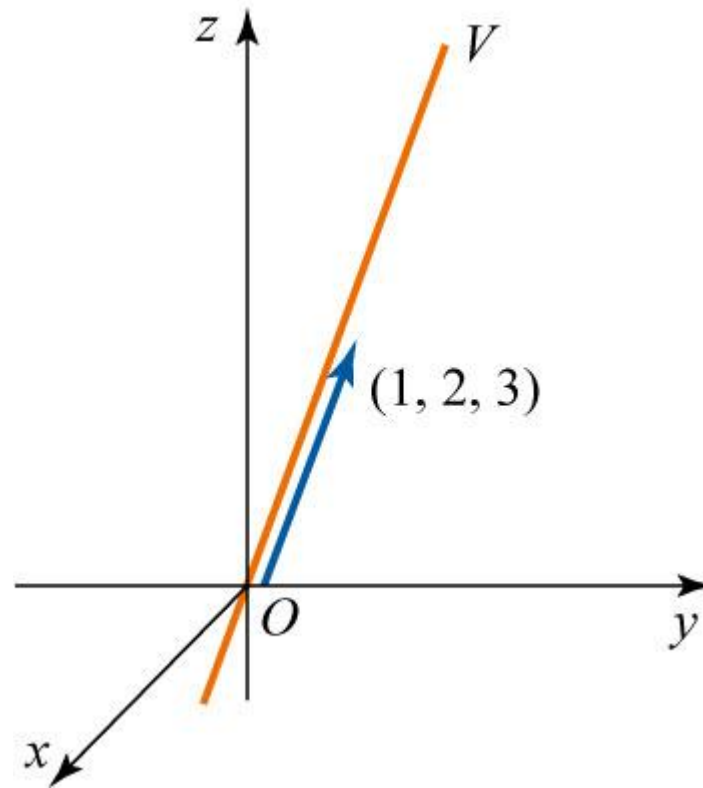
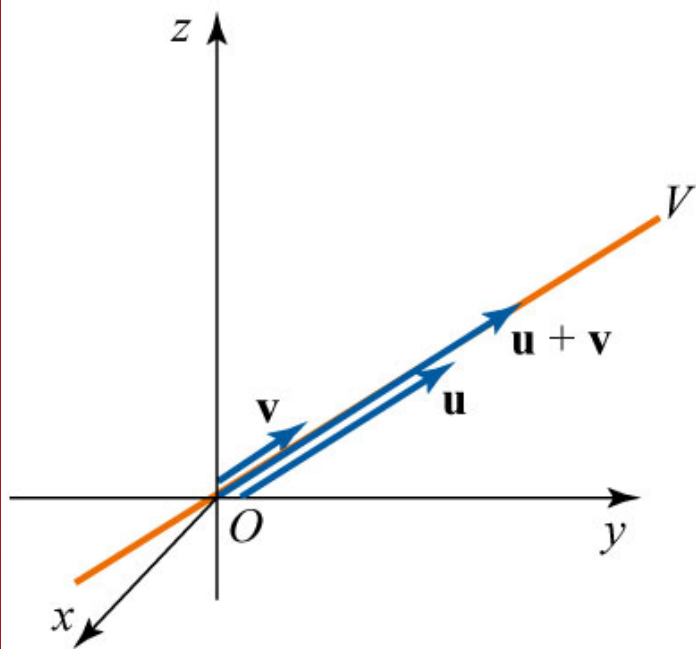
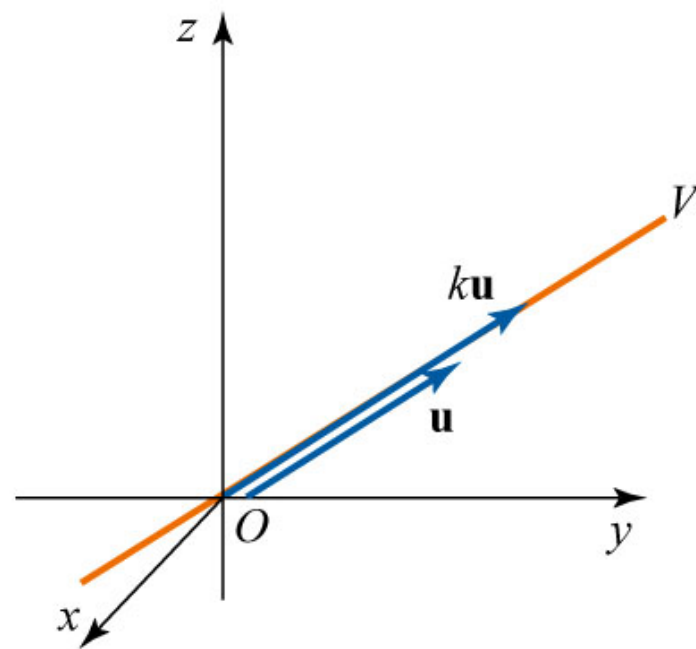


Figure 1.15



V is closed under addition.



V is closed under scalar multiplication.

Figure 1.16

Example 3

Let W be the set of vectors of the form (a, a^2, b) . Show that W is not a subspace of $\mathbf{R}^3$.

Solution

W consists of all elements of $\mathbf{R}^3$ for which the second component is the square of the first. For example, the vector $(2, 4, 3)$ is in W , whereas the vector $(2, 5, 3)$ is not. Let us check for closure under addition. Let (a, a^2, b) and (c, c^2, d) be elements of W . We get

$$\begin{aligned}(a, a^2, b) + (c, c^2, d) &= (a + c, a^2 + c^2, b + d) \\ &\neq (a + c, (a + c)^2, b + d)\end{aligned}$$

Thus $(a, a^2, b) + (c, c^2, d)$ is not an element of W . The set W is not closed under addition. Thus, W is not a subspace. This completes the proof that W is not a subspace. Let us, however, illustrate the check for closure under scalar multiplication. Let k be a scalar. We get

$$\begin{aligned} k(a, a^2, b) &= (ka, ka^2, kb) \\ &\neq (ka, (ka)^2, kb) \end{aligned}$$

Thus $k(a, a^2, b)$ is not an element of W . W is not closed under scalar multiplication either.

Example 4

Consider the following homogeneous system of linear equations:

$$x_1 - x_2 + 3x_3 = 0$$

$$x_2 - 5x_3 = 0$$

$$2x_1 - x_2 + x_3 = 0$$

It can be shown that there are many solutions $x_1 = 2r$, $x_2 = 5r$, $x_3 = r$. We can write these solutions as vectors in $\mathbf{R}^3$ as follows

$$(2r, 5r, r)$$

The set of solutions W thus consists of vectors for which the first component is twice the third, and the second component is five times the third component. The vector $(4, 10, 2)$ for example is in W while $(4, 9, 2)$ is not. Let us show that W is closed under addition and under scalar multiplication. Let $\mathbf{u} = (2r, 5r, r)$ and $\mathbf{v} = (2s, 5s, s)$ be arbitrary vectors in W . We get

$$\begin{aligned}\mathbf{u} + \mathbf{v} &= (2r + 2s, 5r + 5s, r + s) \\ &= (2(r + s), 5(r + s), r + s)\end{aligned}$$

and

$$\begin{aligned}k\mathbf{u} &= k(2r, 5r, r) \\ &= (2kr, 5kr, kr)\end{aligned}$$

In $\mathbf{u} + \mathbf{v}$ and $k\mathbf{u}$ the first component is twice the third, and the second component is five times the third. They are both in W . W is closed under addition and under scalar multiplication.

It is a subspace of $\mathbf{R}^3$. Writing the vector $(2r, 5r, r)$ in the form $r(2, 5, 1)$ we see that the set of solutions is the line defined by the vector $(2, 5, 1)$. See Figure 1.17. The vector $(2, 5, 1)$ spans the set of solutions.

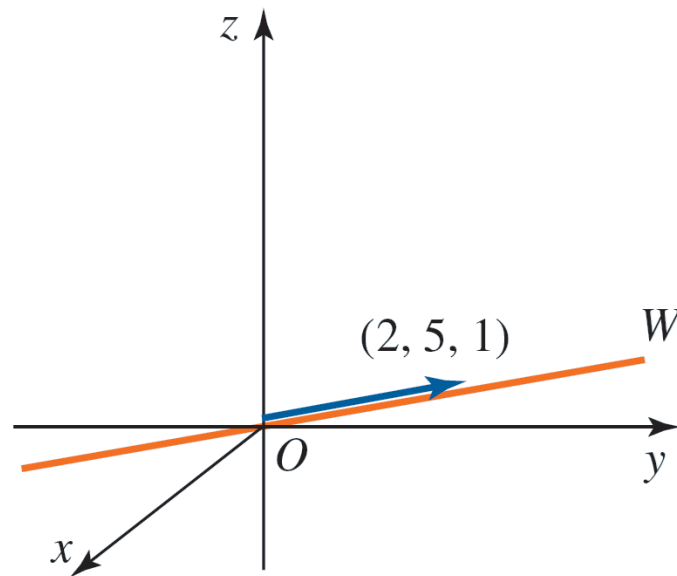


Figure 1.17

Example 5

Consider the following system of homogeneous linear equations.

$$x_1 - x_2 + x_3 + 2x_4 = 0$$

$$x_1 - 3x_3 + 2x_4 = 0$$

$$2x_1 - x_2 - 2x_3 + 4x_4 = 0$$

It can be shown that there are many solutions $x_1 = 3r - 2s$, $x_2 = 4r$, $x_3 = r$, $x_4 = s$. Write these solutions as vectors in $\mathbf{R}^4$,

$$(3r - 2s, 4r, r, s)$$

It can be shown that this set W of vectors is closed under addition and scalar multiplication, and is thus a subspace of $\mathbf{R}^4$. Separate the variables in the general solution.

$$(3r - 2s, 4r, r, s) = r(3, 4, 1, 0) + s(-2, 0, 0, 1)$$

This implies that every vector in W can be expressed as a linear combination of $(3, 4, 1, 0)$ and $(-2, 0, 0, 1)$. The vectors $(3, 4, 1, 0)$ and $(-2, 0, 0, 1)$ span W . W will be a plane through the origin in a four-space.

We shall see in Section 2.2 that the set of solutions to every homogeneous system of linear equations is a subspace. An implication of this result is that the sum of any two solutions is a solution. The scalar multiple of a solution is also a solution. However, we shall find that the set of solutions to a nonhomogeneous system is not a subspace. We shall discuss ways of relating solutions of nonhomogeneous systems to solutions of “corresponding” systems of homogeneous linear equations.

1.5 Basis and Dimension in $\mathbf{R}^n$

Standard Basis of $\mathbf{R}^n$

Consider the vectors $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ in $\mathbf{R}^3$.

These vectors have two very important properties:

- (i) They *span* $\mathbf{R}^3$. That is, we can write an arbitrary vector (x, y, z) as a linear combination of the vectors as follows:

$$(x, y, z) = x(1, 0, 0) + y(0, 1, 0) + z(0, 0, 1)$$

- (ii) They are *linearly independent*. That is, the identity

$$p(1, 0, 0) + q(0, 1, 0) + r(0, 0, 1) = (0, 0, 0)$$

for scalars p , q , and r has the unique solution $p = 0$, $q = 0$, $r = 0$.

Dimension of $\mathbf{R}^n$

A finite set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is called a **basis** for a vector space V if the set spans V and is linearly independent.

The set of n vectors

$$\{(1, 0, \dots, 0), (0, 1, \dots, 0), \dots, (0, \dots, 1)\}$$

is a *basis* for $\mathbf{R}^n$. This basis is called the **standard basis** for $\mathbf{R}^n$.

The set $\{(1, 0, \dots, 0), (0, 1, \dots, 0), \dots, (0, 0, \dots, 1)\}$ of n vector is the standard basis of $\mathbf{R}^n$. The dimension of $\mathbf{R}^n$ is n .

Span, Linear Independence, and Basis

- ◆ **Span** The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ span a space if every vector $\mathbf{v}$ in the space can be expressed as a linear combination of them,
$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n.$$
- ◆ **Linear Independence** The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent if the identity $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0}$ is only true for $c_1=0, c_2=0, \dots, c_n=0$. If there are nonzero values of $c_1, c_2, \dots, c_n$ for which this identity holds, the set is said to be linearly dependent.
- ◆ **Basis and Dimension** A set of vectors is a basis for a space if the vectors (i) span the space and (ii) are linearly independent. The number of vectors in a basis is unique. This number is called the dimension of the space.

Example 1

Consider the subspace W of $\mathbf{R}^3$ of vectors of the form $(a, b, a + b)$ that was discussed in the previous section. Let $\mathbf{u} = (a, b, a + b)$ be an arbitrary vector in W . We know that we can write $\mathbf{u}$ in the form

$$\mathbf{u} = a(1, 0, 1) + b(0, 1, 1)$$

The vector $(1, 0, 1)$ and $(0, 1, 1)$ thus span W . Further, the identity $p(1, 0, 1) + q(0, 1, 1) = (0, 0, 0)$ leads to $p = 0, q = 0$. These vectors are linear independent. The set $\{(1, 0, 1), (0, 1, 1)\}$ is a basis for W . The dimension of W is 2. We know that W is a plane through the origin, fitting in with our intuitive thinking of a plane as being a two-dimensional space. See Figure 1.18.

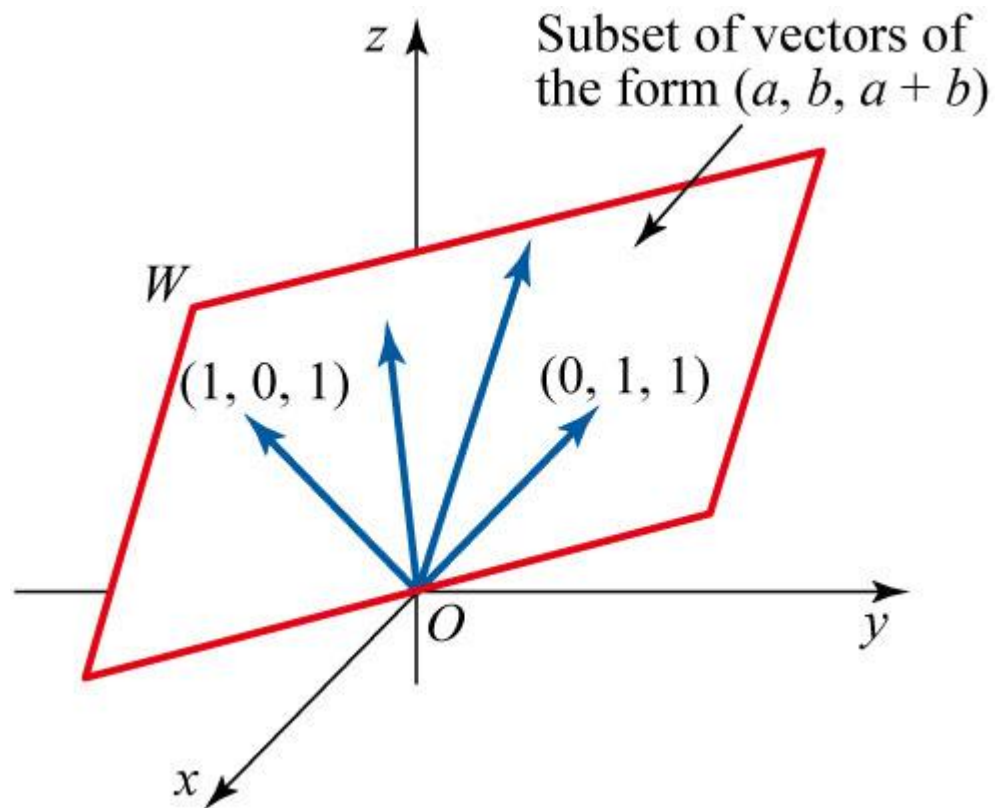


Figure 1.18

Example 2

Consider the subset V of $\mathbf{R}^2$ of vectors of the form $(a, 2a)$, where the second component is twice the first component. Let us show that V is a subspace. Let $\mathbf{u} = (a, 2a)$ and $\mathbf{v} = (b, 2b)$ be vectors in V and k be a scalar. Then

$$\begin{aligned}\mathbf{u} + \mathbf{v} &= (a, 2a) + (b, 2b) \\ &= (a + b, 2a + 2b) \\ &= (a + b, 2(a + b))\end{aligned}$$

and

$$\begin{aligned}k\mathbf{u} &= k(a, 2a) \\ &= (ka, 2ka)\end{aligned}$$

Both $\mathbf{u} + \mathbf{v}$ and $k\mathbf{u}$ are in V since their second components are twice the first. V is closed under addition and scalar multiplication. It is a subspace of $\mathbf{R}^2$; it is a vector space in its own right.

We now proceed to find a basis for V . Write the arbitrary vector as follows:

$$(a, 2a) = a(1, 2)$$

The elements of V are scalar multiples of the single vector $(1, 2)$. The implication is that the set $\{(1, 2)\}$ is a basis for V . The dimension of V is 1. The vectors form the line through the origin defined by $(1, 2)$. See Figure 1.19.

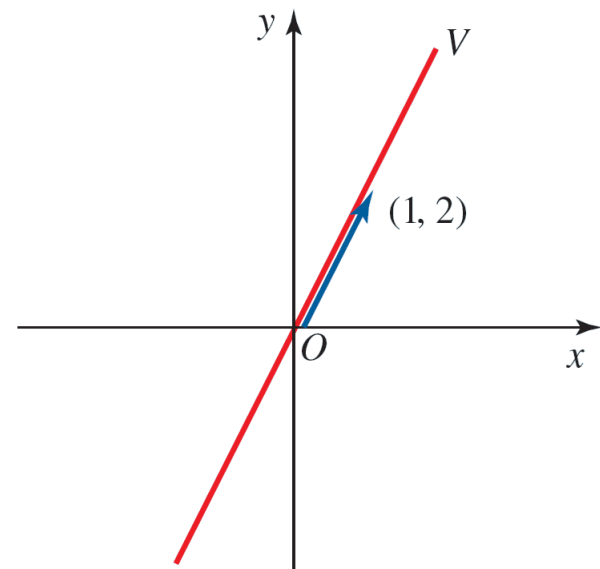


Figure 1.19

Example 3

Consider the following homogeneous system of linear equation:

$$x_1 + x_2 - 7x_3 + 2x_4 + x_5 = 0$$

$$x_1 + 2x_2 - 10x_3 + 2x_4 + 2x_5 = 0$$

$$2x_1 + 3x_2 - 17x_3 + 4x_4 + 3x_5 = 0$$

It can be shown that the general solution to this system in vector form is $(4r - 2s, 3r - t, r, s, t)$ and that this set is closed under addition and under scalar multiplication. It is a subspace W of $\mathbf{R}^5$. Let us find a basis and the dimension of this subspace. Write

$$(4r - 2s, 3r - t, r, s, t)$$

$$= r(4, 3, 1, 0, 0) + s(-2, 0, 0, 1, 0) + t(0, -1, 0, 0, 1)$$

The vectors $(4, 3, 1, 0, 0)$, $(-2, 0, 0, 1, 0)$, $(0, -1, 0, 0, 1)$ span W . Let us check for linear independence. Consider the following identity for arbitrary variables p, q, h .

$$p(4, 3, 1, 0, 0) + q(-2, 0, 0, 1, 0) + h(0, -1, 0, 0, 1) = (0, 0, 0, 0, 0)$$

We get

$$(4p - 2q, 3p - h, p, q, h) = (0, 0, 0, 0, 0)$$

This implies that $p = 0, q = 0, h = 0$. thus, the vectors are linearly independent.

The set $\{(4, 3, 1, 0, 0), (-2, 0, 0, 1, 0), (0, -1, 0, 0, 1)\}$ is a basis for W . the set of solution is a three-dimensional vector space embedded in a five-dimensional space.

Example 4

- (i) Determine whether the following sets of vectors are linear dependent or independent in $\mathbf{R}^3$. (a) $\{(1, 2, 0), (0, 1, -1), (1, 1, 2)\}$.
(b) $\{(1, 2, 3), (-2, 0, 1), (4, -4, -9)\}$.

Solution

$$c_1(1, 2, 0) + c_2(0, 1, -1) + c_3(1, 1, 2) = (0, 0, 0)$$

$$c_1 + c_3 = 0$$

$$2c_1 + c_2 + c_3 = 0$$

$$-c_2 + 2c_3 = 0$$

The system has the unique solution $c_1 = 0$, $c_2 = 0$, $c_3 = 0$. Thus, the vectors are linearly independent.

1.6 Dot Product, Norm, Angle, and Distance

Definition

Let $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$ be two vectors in $\mathbf{R}^n$. The **dot product** of $\mathbf{u}$ and $\mathbf{v}$ is denoted $\mathbf{u} \cdot \mathbf{v}$ and is defined by

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + \cdots + u_n v_n$$

The dot product assigns a real number to each pair of vectors.

For example, if $\mathbf{u} = (1, -2, 4)$ and $\mathbf{v} = (3, 0, 2)$ their dot product is

$$\begin{aligned}\mathbf{u} \cdot \mathbf{v} &= (1 \times 3) + (-2 \times 0) + (4 \times 2) \\ &= 3 + 0 + 8 \\ &= 11\end{aligned}$$

Properties of the Dot Product

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in $\mathbf{R}^n$ and let c be a scalar. Then

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$
3. $c\mathbf{u} \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot c\mathbf{v}$
4. $\mathbf{u} \cdot \mathbf{u} \geq 0$, and $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$

Proof

1. Let $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$. We get

$$\begin{aligned}\mathbf{u} \cdot \mathbf{v} &= u_1 v_1 + \cdots + u_n v_n \\ &= v_1 u_1 + \cdots + v_n u_n \\ &= \mathbf{v} \cdot \mathbf{u}\end{aligned}$$

by the commutative property of real numbers

4. $\mathbf{u} \cdot \mathbf{u} \geq 0$, and $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$

Proof

$$\mathbf{u} \cdot \mathbf{u} = u_1 u_1 + \cdots + u_n u_n = (u_1)^2 + \cdots + (u_n)^2.$$

$(u_1)^2 + \cdots + (u_n)^2 \geq 0$, since it is the sum of squares.

Thus $\mathbf{u} \cdot \mathbf{u} \geq 0$.

Further, $(u_1)^2 + \cdots + (u_n)^2 = 0$ if and only if

$u_1 = 0, \dots, u_n = 0$. Thus $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$

Norm of a Vector in $\mathbb{R}^n$

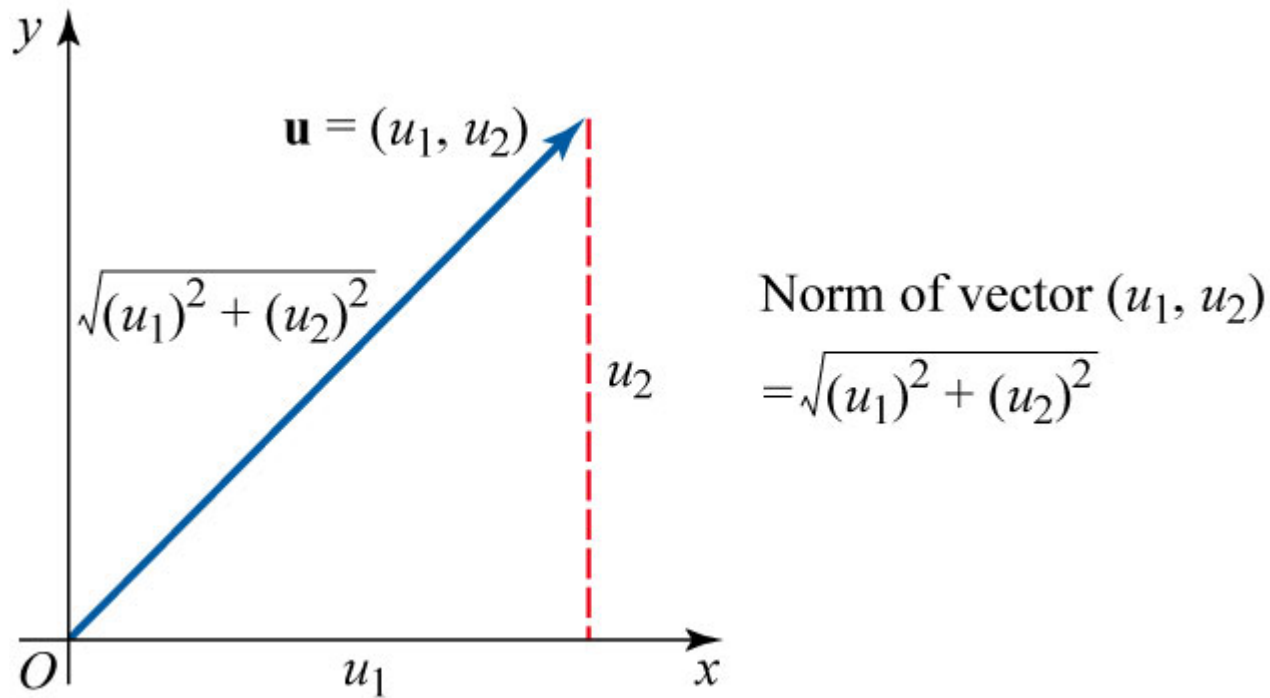


Figure 1.20

Definition

The **norm** (**length** or **magnitude**) of a vector $\mathbf{u} = (u_1, \dots, u_n)$ in $\mathbf{R}^n$ is denoted $\|\mathbf{u}\|$ and defined by

$$\|\mathbf{u}\| = \sqrt{(u_1)^2 + \dots + (u_n)^2}$$

Note: The norm of a vector can also be written in terms of the dot product

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}$$

For example, if $\mathbf{u} = (1, 3, 5)$ in $\mathbf{R}^3$ and $\mathbf{v} = (3, 0, 1, 4)$ of $\mathbf{R}^4$, then

$$\|\mathbf{u}\| = \sqrt{(1)^2 + (3)^2 + (5)^2} = \sqrt{1+9+25} = \sqrt{35}$$

$$\|\mathbf{v}\| = \sqrt{(3)^2 + (0)^2 + (1)^2 + (4)^2} = \sqrt{9+0+1+16} = \sqrt{26}$$

Definition

A **unit vector** is a vector whose norm is 1. If $\mathbf{v}$ is a nonzero vector, then the vector

$$\mathbf{u} = \frac{1}{\|\mathbf{v}\|} \mathbf{v}$$

is a unit vector in the direction of $\mathbf{v}$.

This procedure of constructing a unit vector in the same direction as a given vector is called **normalizing** the vector.

Example 1

Find the norm of the vector $(2, -1, 3)$. Normalize this vector.

Solution

$\|(2, -1, 3)\| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{14}$. The norm of $(2, -1, 3)$ is $\sqrt{14}$.

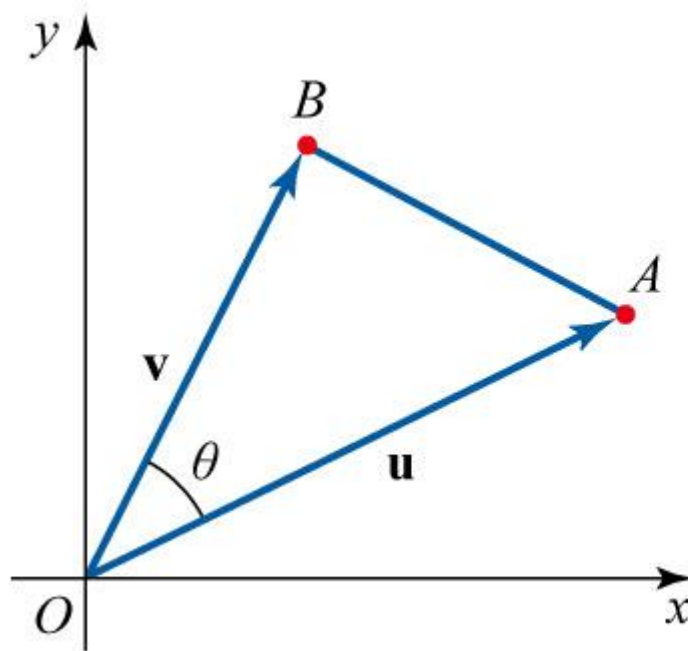
The normalized vector is

$$\frac{1}{\sqrt{14}}(2, -1, 3)$$

The vector may also be written $\left(\frac{2}{\sqrt{14}}, \frac{-1}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right)$.

This vector is a unit vector in the direction of $(2, -1, 3)$.

Algebra Between Vectors



$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

Figure 1.21

The Cauchy-Schwartz Inequality

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbf{R}^n$, then

$$\left| \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \right| \leq 1$$

Here $|\mathbf{u} \cdot \mathbf{v}|$ denotes the absolute value of the number $\mathbf{u} \cdot \mathbf{v}$.

Definition

Let $\mathbf{u}$ and $\mathbf{v}$ be two nonzero vectors in $\mathbf{R}^n$. The cosine of the angle θ between these vectors is

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \quad 0 \leq \theta \leq \pi$$

Example 2

Determine the angle between the vectors $\mathbf{u} = (1, 0, 0)$ and $\mathbf{v} = (1, 0, 1)$ in $\mathbf{R}^3$.

Solution

$$\mathbf{u} \cdot \mathbf{v} = (1, 0, 0) \cdot (1, 0, 1) = 1$$

$$\|\mathbf{u}\| = \sqrt{1^2 + 0^2 + 0^2} = 1 \quad \|\mathbf{v}\| = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}$$

$$\text{Thus, } \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{1}{\sqrt{2}},$$

The angle between $\mathbf{u}$ and $\mathbf{v}$ is $\pi/4$ (or 45°).

THEOREM 1.4

Two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if

$$\mathbf{u} \cdot \mathbf{v} = 0.$$

For example, the vector $(2, -3, 1)$ and $(1, 2, 4)$ are orthogonal since

$$\begin{aligned} &(2, -3, 1) \cdot (1, 2, 4) \\ &= (2 \times 1) + (-3 \times 2) + (1 \times 4) \\ &= 2 - 6 + 4 \\ &= 0. \end{aligned}$$

Properties of Standard Basis of $\mathbf{R}^n$

The standard basis for $\mathbf{R}^n$, $\{(1, 0, \dots, 0), (0, 1, \dots, 0), \dots, (0, 0, \dots, 1)\}$, is an orthonormal set.

Example 3

- (a) Let $\mathbf{w}$ be a vector in $\mathbf{R}^n$. Let W be the set of vectors that are orthogonal to $\mathbf{w}$. Show that W is a subspace of $\mathbf{R}^n$.
- (b) Find a basis for the subspace W of vectors that are orthogonal to $\mathbf{w} = (1, 3, 1)$. Give the dimension and a geometrical description of W .

Solution

- (a) Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in W . Each vector is orthogonal to $\mathbf{w}$. Thus and Using the properties of dot product,

$$\mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w} = 0$$

$$(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = 0$$

Thus $\mathbf{u} + \mathbf{v}$ is orthogonal to $\mathbf{w}$, implying that $\mathbf{u} + \mathbf{v}$ is in W . W is closed under addition. Further, if c is a scalar,

$$c(\mathbf{u} \cdot \mathbf{w}) = 0$$

$$c\mathbf{u} \cdot \mathbf{w} = 0$$

This means that the vector $c\mathbf{u}$ is orthogonal to $\mathbf{w}$ and hence is in W . W is closed under scalar multiplication. W is thus a subspace of $\mathbf{R}^n$.

Solution

(b) W is the set of vectors in $\mathbf{R}^3$ that are orthogonal to $\mathbf{w} = (1, 3, 1)$. Let (a, b, c) be a vector in W . Thus

$$(a, b, c) \cdot (1, 3, 1) = 0$$

$$a + 3b + c = 0$$

$$c = -a - 3b$$

Therefore W is the set of vectors of the form $(a, b, -a - 3b)$. Separate the variables a and b ,

$$(a, b, -a - 3b) = a(1, 0, -1) + b(0, 1, -3)$$

Every vector in W can be written as a linear combination of $(1, 0, -1)$ and $(0, 1, -3)$. Thus the vectors span W . It can be shown that these vectors are also linearly independent. The set $\{(1, 0, -1), (0, 1, -3)\}$ is therefore a basis for W .

Now for the geometry. Each vector in W is of the form $a(1, 0, -1) + b(0, 1, -3)$. As a and b vary we get the various vectors that make up W . When $a = 1$ and $b = 2$ for example, we get the vector $(1, 2, -7)$. Since there are two vectors in the basis the dimension of W is 2. A subspace of dimension 2 is a plane. Thus W is the plane in $\mathbf{R}^3$ defined by $(1, 0, -1)$ and $(0, 1, -3)$. See Figure 1.22.

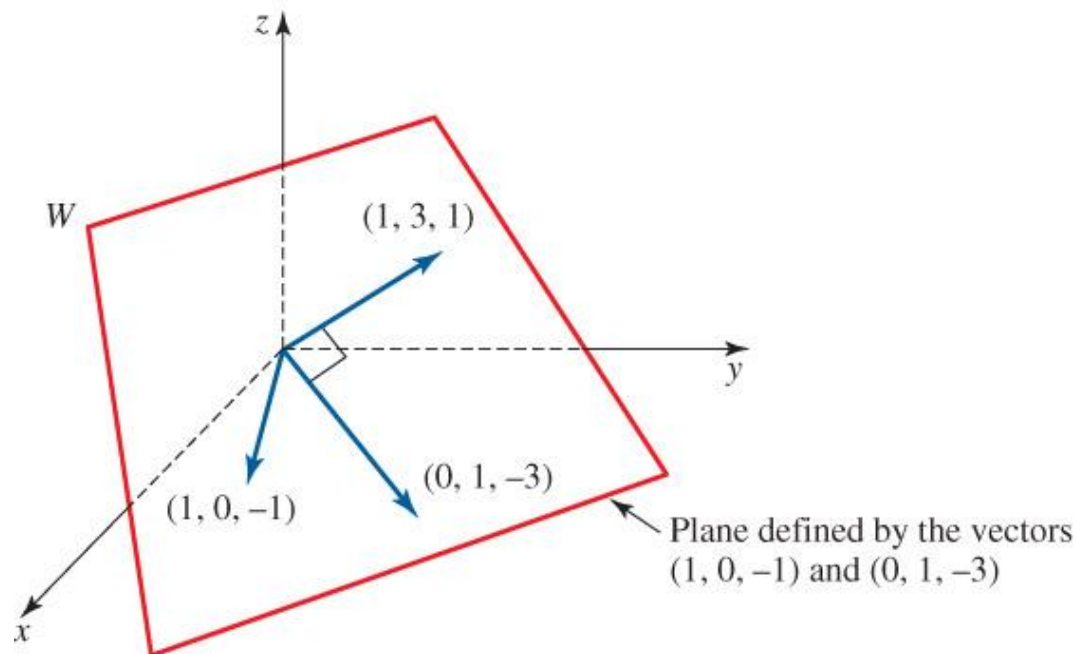


Figure 1.22

Example 4

- (a) Show that the following vectors are orthogonal.
(b) Compute the norm of each vector.

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -5 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}$$

Solution

- (a) The dot product of $\mathbf{u}$ and $\mathbf{v}$ is obtained by multiplying corresponding components and adding.

$$\mathbf{u} \cdot \mathbf{v} = (1 \times -1) + (2 \times 3) + (-5 \times 1) = 0$$

The dot product is zero, thus the vectors are orthogonal.

Solution

(b) The norm of each vector is obtained by taking the square root of the sum of the squares of the components.

$$\|\mathbf{u}\| = \sqrt{1^2 + 2^2 + (-5)^2} = \sqrt{30}$$

$$\|\mathbf{v}\| = \sqrt{(-1)^2 + 3^2 + 1^2} = \sqrt{11}$$

Distance between Points

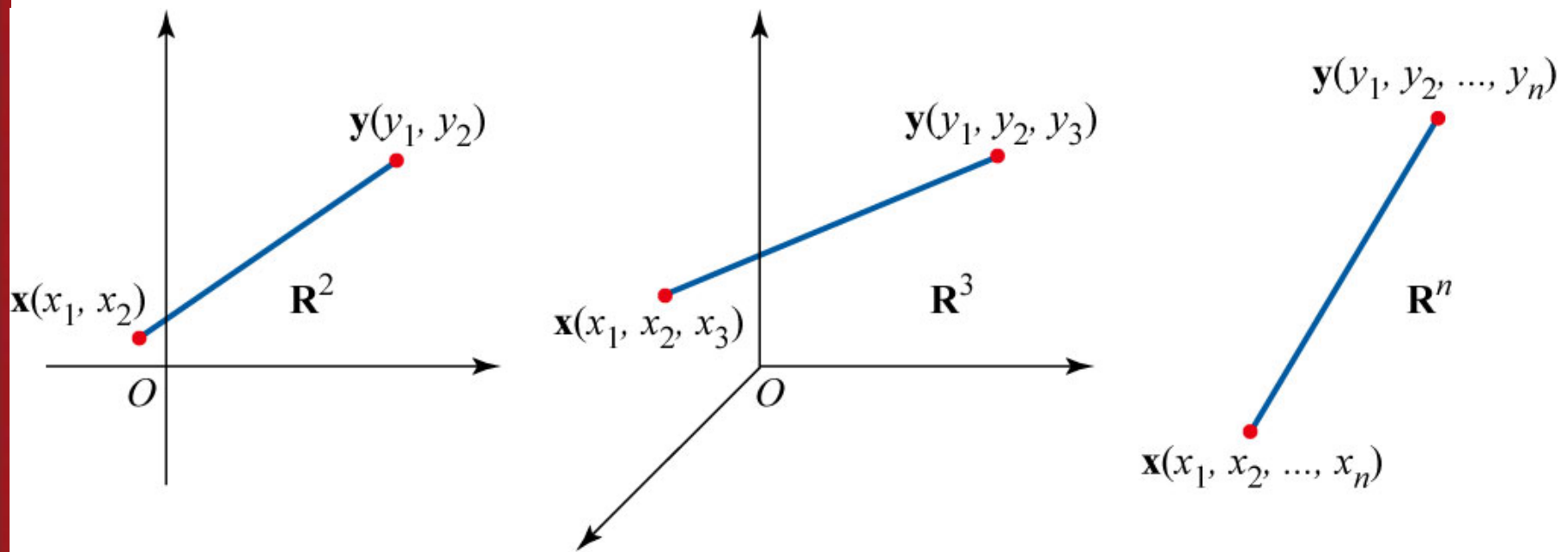
Definition

Let $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)$ be two points in $\mathbf{R}^n$. The **distance between $\mathbf{x}$ and $\mathbf{y}$** is denoted $d(\mathbf{x}, \mathbf{y})$ and is defined by

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2}$$

Note: We can also write this distance formula as

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$$



$$\mathbf{R}^2: d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

$$\mathbf{R}^3: d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2}$$

$$\mathbf{R}^n: d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2}$$

Figure 1.23

Example 5

Determine the distance between the points $\mathbf{x} = (1, -2, 3, 0)$ and $\mathbf{y} = (4, 0, -3, 5)$ in $\mathbf{R}^4$.

Solution

$$\begin{aligned}d(\mathbf{x} \cdot \mathbf{y}) &= \sqrt{(1-4)^2 + (-2-0)^2 + (3+3)^2 + (0-5)^2} \\&= \sqrt{9 + 4 + 36 + 25} \\&= \sqrt{74}\end{aligned}$$

Example 6

Prove that the distance in $\mathbf{R}^n$ has the following **symmetric property**: $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$.

Solution

Let $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)$

We get

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) &= \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2} \\ &= \sqrt{(y_1 - x_1)^2 + \dots + (y_n - x_n)^2} \\ &= d(\mathbf{y}, \mathbf{x}) \end{aligned}$$

Euclidean Geometry of $\mathbb{R}^n$

• Dot product of vectors $\mathbf{u}$ and $\mathbf{v}$: $\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + \cdots + u_n v_n$

• Norm of a vector $\mathbf{u}$: $\|\mathbf{u}\| = \sqrt{(u_1)^2 + \cdots + (u_n)^2}$

• Angle between vectors $\mathbf{u}$ and $\mathbf{v}$: $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$

• Distance between points $\mathbf{x}$ and $\mathbf{y}$: $d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}$

Properties of Norm

1. $\|\mathbf{u}\| \geq 0$
(the length of a vector cannot be negative)
2. $\|\mathbf{u}\| = 0$ if and only if $\mathbf{u} = \mathbf{0}$
(the length of a vector is zero if and only if the vector is the zero vector)
3. $\|c\mathbf{u}\| = |c|\|\mathbf{u}\|$
(the length of $c\mathbf{u}$ is $|c|$ times the length of $\mathbf{u}$.)

Properties of Distance

1. $d(\mathbf{x}, \mathbf{y}) \geq 0$
(the distance between two points cannot be negative)
2. $d(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x} = \mathbf{y}$
(the distance between two points is zero if and only if the points are coincident)
3. $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ Symmetry property)
(the distance between $\mathbf{x}$ and $\mathbf{y}$ is the same as the distance between $\mathbf{y}$ and $\mathbf{x}$)

THEOREM 1.5

The Cauchy-Schwartz Inequality

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbf{R}^n$, then

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

Here $|\mathbf{u} \cdot \mathbf{v}|$ denotes the absolute value of the number $\mathbf{u} \cdot \mathbf{v}$.

Proof

We first look at the special case when $\mathbf{u} = \mathbf{0}$. If $\mathbf{u} = \mathbf{0}$ then $|\mathbf{u} \cdot \mathbf{v}| = 0$ and $\|\mathbf{u}\| \cdot \|\mathbf{v}\| = 0$. Equality holds.

Let us now consider $\mathbf{u} \neq \mathbf{0}$. The proof of the inequality for this case involves a very interesting application of the quadratic formula. Consider the vector $r\mathbf{u} + \mathbf{v}$, where r is any real number. Using the properties of the dot product we get

$$(r\mathbf{u} + \mathbf{v}) \cdot (r\mathbf{u} + \mathbf{v}) = r^2(\mathbf{u} \cdot \mathbf{u}) + 2r(\mathbf{u} \cdot \mathbf{v}) + \mathbf{v} \cdot \mathbf{v}$$

Proof cont.

The properties of the dot product also tell us that

$$(r\mathbf{u} + \mathbf{v}) \cdot (r\mathbf{u} + \mathbf{v}) \geq 0$$

Combining these results we get

$$r^2(\mathbf{u} \cdot \mathbf{u}) + 2r(\mathbf{u} \cdot \mathbf{v}) + \mathbf{v} \cdot \mathbf{v} \geq 0$$

Let $a = \mathbf{u} \cdot \mathbf{u}$, $b = 2\mathbf{u} \cdot \mathbf{v}$, $c = \mathbf{v} \cdot \mathbf{v}$. Thus

$$ar^2 + br + c \geq 0$$

This implies that the **quadratic function** $f(r) = ar^2 + br + c$ is never negative. Therefore $f(r)$ whose graph is a parabola, must have either one zero or no zeros. The zeros of $f(r)$ are the roots of the equation

$$ar^2 + br + c = 0$$

Proof cont.

The **discriminant** gives information about these zeros. There is one zero if $b^2 - 4ac = 0$ and no zeros if $b^2 - 4ac < 0$. Thus

$$b^2 - 4ac \leq 0$$

$$b^2 \leq 4ac$$

$$(2\mathbf{u} \cdot \mathbf{v})^2 \leq 4(\mathbf{u} \cdot \mathbf{u})(\mathbf{v} \cdot \mathbf{v})$$

$$(\mathbf{u} \cdot \mathbf{v})^2 \leq (\mathbf{u} \cdot \mathbf{u})(\mathbf{v} \cdot \mathbf{v})$$

$$(\mathbf{u} \cdot \mathbf{v})^2 \leq \|\mathbf{u}\|^2 \|\mathbf{v}\|^2$$

Taking the square root of both sides gives the Cauchy-Schwarz inequality.

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

THEOREM 1.6

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in $\mathbf{R}^n$.

(a) Triangle inequality: $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

The length of one side of a triangle cannot exceed the sum of the lengths of the other two sides.

(a) Pythagorean theorem: If $\mathbf{u} \cdot \mathbf{v} = 0$, then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.
The square of the hypotenuse of a right triangle is equal to the sum of the squares of the other two sides.

Proof (a)

(a) By the properties of dot product and norm,

$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 &= (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) \\&= \mathbf{u} \cdot \mathbf{u} + \mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{u} + \mathbf{v} \cdot \mathbf{v} \\&= \mathbf{u} \cdot \mathbf{u} + 2\mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{v} \\&= \|\mathbf{u}\|^2 + 2\mathbf{u} \cdot \mathbf{v} + \|\mathbf{v}\|^2 \\&\leq \|\mathbf{u}\|^2 + 2|\mathbf{u} \cdot \mathbf{v}| + \|\mathbf{v}\|^2\end{aligned}$$

Proof (a) cont.

The Cauchy-Schwarz inequality implies that

$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 &\leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 \\ &= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2\end{aligned}$$

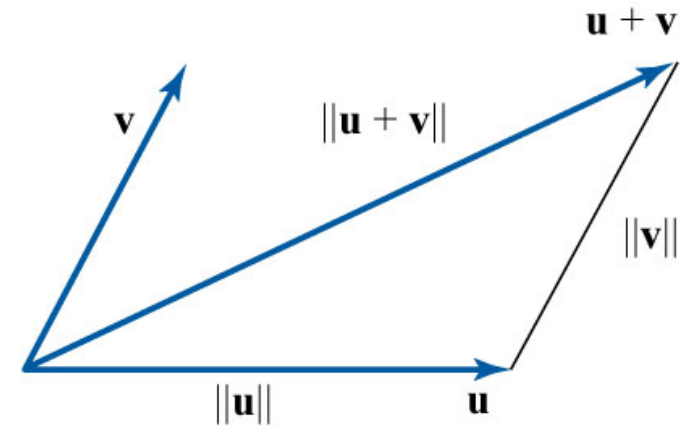
Taking the square root of each side gives the triangle inequality.

Proof (b)

(b) By the properties of dot product, and the fact that $\mathbf{u} \cdot \mathbf{v} = 0$, we get

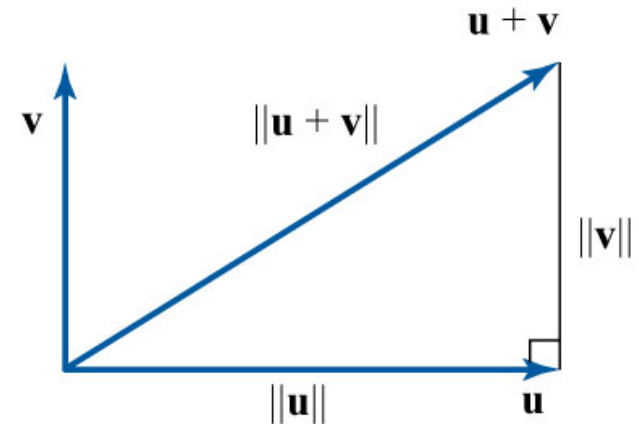
$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 &= (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) \\ &= \mathbf{u} \cdot \mathbf{u} + 2\mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{v} \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\end{aligned}$$

proving the Pythagorean Theorem.



$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$
Triangle inequality

(a)



$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$
Pythagorean Theorem

(b)

Figure 1.24

1.7 Curve Fitting, Electrical Networks, and Traffic Flow

Curve Fitting

- A set of data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ is given and it is necessary to find a polynomial whose graph passes through the points.
- The points are often measurements in an experiment.
- The x-coordinates are called **base points**.
- It can be shown that if the base points are all distinct, then a unique polynomial of degree $n - 1$ (or less)

$$y = a_0 + a_1x + \dots + a_{n-2}x^{n-2} + a_{n-1}x^{n-1}$$

can be **fitted** to the points. See Figure 1.25.

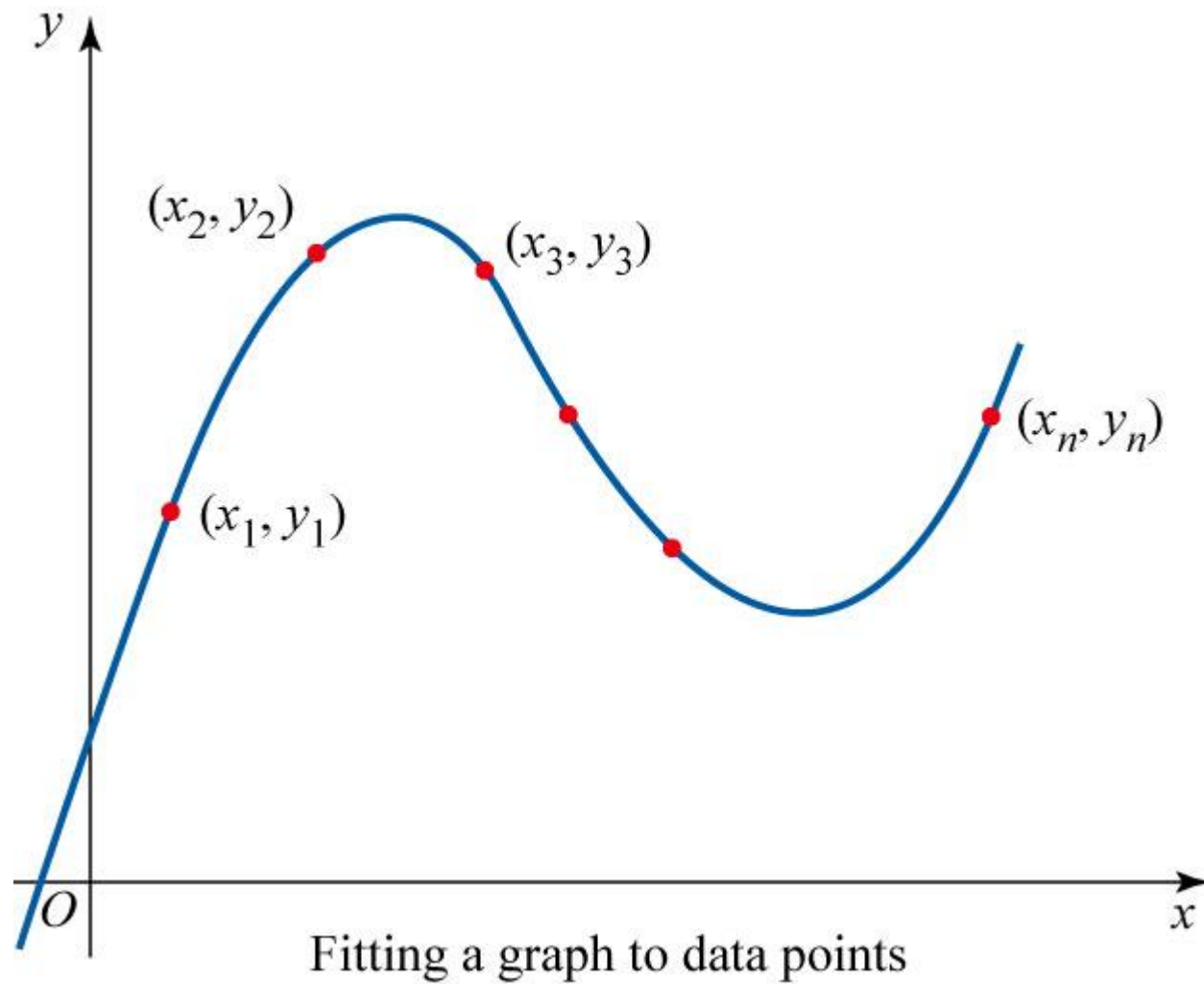


Figure 1.25

Example 1

Determine the equation of the polynomial of degree two whose graph passes through the points (1, 6), (2, 3), (3, 2).

Solution

Let the polynomial be $y = a_0 + a_1x + a_2x^2$, and substituting $x = 1, y = 6$; $x = 2, y = 3$; $x = 3, y = 2$, we have

$$a_0 + a_1 + a_2 = 3$$

$$a_0 + 2a_1 + 4a_2 = 3$$

$$a_0 + 3a_1 + 9a_2 = 2$$

Solving this system for a_0 , a_1 , and a_2 using Gauss-Jordan elimination.

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 4 & 3 \\ 1 & 3 & 9 & 2 \end{bmatrix} \xrightarrow[\text{R3} + (-1)\text{R1}]{\text{R2} + (-1)\text{R1}} \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 3 & -3 \\ 0 & 2 & 8 & -4 \end{bmatrix} \xrightarrow[\text{R3} + (-1)\text{R2}]{\text{R1} + (-1)\text{R2}} \begin{bmatrix} 1 & 0 & -2 & 9 \\ 0 & 1 & 3 & -3 \\ 0 & 0 & 2 & 2 \end{bmatrix}$$

$$\begin{array}{l} \approx \\ \left(\frac{1}{2}\right)R_3 \end{array} \begin{bmatrix} 1 & 0 & -2 & 9 \\ 0 & 1 & 3 & -3 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{array}{l} \approx \\ R_1 + 2R_3 \\ R_2 + (-3)R_3 \end{array} \begin{bmatrix} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & -6 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

We get $a_0 = 11$, $a_1 = -6$, $a_2 = 1$.

The parabola that passes through these points is $y = 11 - 6x + x^2$.
Such as Figure, 1.26.

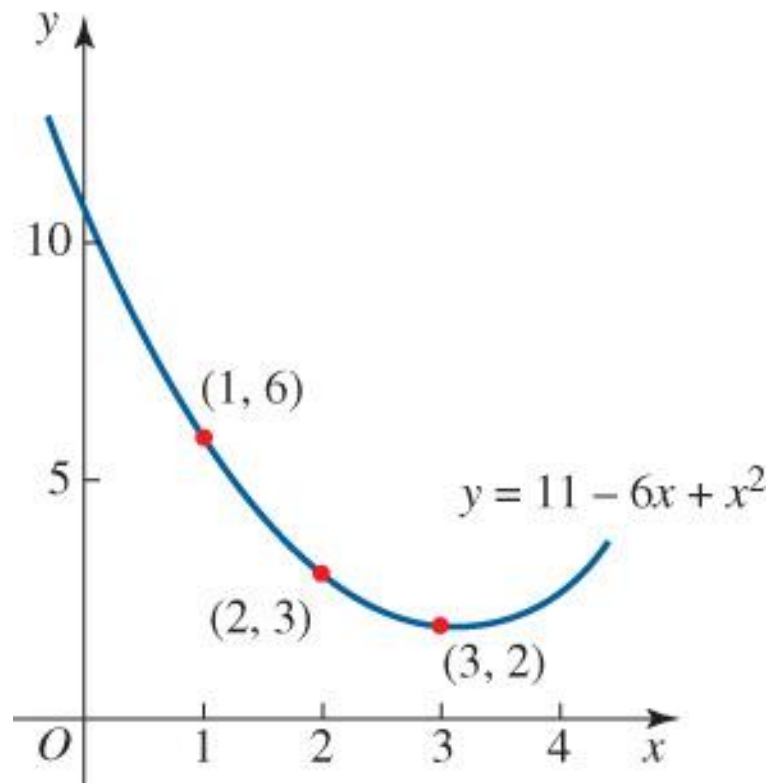


Figure 1.26

Electrical Network Analysis

Kirchhoff's Laws

1. **Junctions:** All the current flowing into a junction must flow out of it. (Current)
2. **Paths:** The sum of the IR terms (I denotes current, R resistance) in any direction around a closed path is equal to the voltage in the path in that direction.

Example 2

Consider the electrical network of Figure 1.27. Let us determine the currents through each branch of this network.

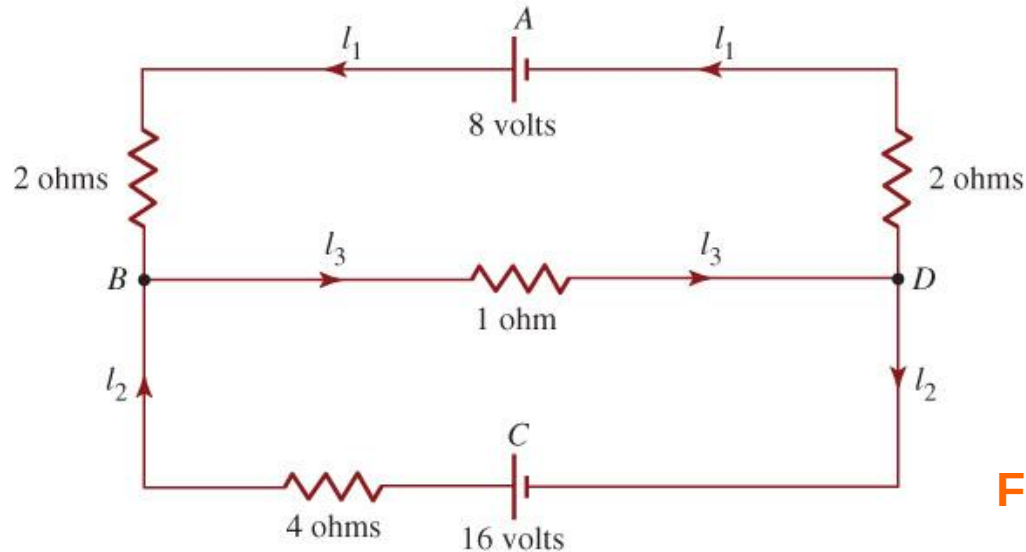


Figure 1.27

Solution

Junctions

$$\text{Junction } B: I_1 + I_2 = I_3$$

$$\text{Junction } D: I_3 = I_1 + I_2$$

These two equations result in a single linear equation

$$I_1 + I_2 - I_3 = 0$$

Paths

$$\text{Path } ABDA: 2I_1 + 1I_3 + 2I_1 = 8$$

$$\text{Path } CBDC: 4I_2 + 1I_3 = 16$$

The problem thus reduces to solving the following system of three linear equations in three variables.

$$\begin{aligned} I_1 + I_2 - I_3 &= 0 \\ 4I_1 + I_3 &= 8 \\ 4I_2 + I_3 &= 16 \end{aligned}$$

Using the method of Gauss-Jordan elimination, we get

$$\begin{aligned} \begin{bmatrix} 1 & 1 & -1 & 0 \\ 4 & 0 & 1 & 8 \\ 0 & 4 & 1 & 16 \end{bmatrix} &\xrightarrow{\text{R2} + (-4)\text{R1}} \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & -4 & 5 & 8 \\ 0 & 4 & 1 & 16 \end{bmatrix} \\ &\approx \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & -\frac{5}{4} & -2 \\ 0 & 4 & 1 & 16 \end{bmatrix} \\ &\xrightarrow{\left(-\frac{1}{4}\right)\text{R2}} \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & -\frac{5}{4} & -2 \\ 0 & 4 & 1 & 16 \end{bmatrix} \end{aligned}$$

$$\begin{array}{l} \approx \\ \text{R1} + (-1)\text{R2} \\ \text{R3} + (-4)\text{R2} \end{array} \begin{bmatrix} 1 & 0 & \frac{1}{4} & 2 \\ 0 & 1 & -\frac{5}{4} & -2 \\ 0 & 0 & 6 & 24 \end{bmatrix} \quad \left(\frac{1}{6}\right) \approx \text{R3} \begin{bmatrix} 1 & 0 & \frac{1}{4} & 2 \\ 0 & 1 & -\frac{5}{4} & -2 \\ 0 & 0 & 1 & 4 \end{bmatrix}$$

$$\approx \begin{array}{l} \text{R1} - \left(\frac{1}{4}\right)\text{R3} \\ \text{R2} + \left(\frac{5}{4}\right)\text{R3} \end{array} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{bmatrix}$$

The currents are $I_1 = 1$, $I_2 = 3$, $I_3 = 4$. The solution is unique.

Example 3

Determine the currents through the various branches of the electrical network in Figure 1.28. This example illustrates how one has to be conscious of direction in applying law 2 for closed paths.

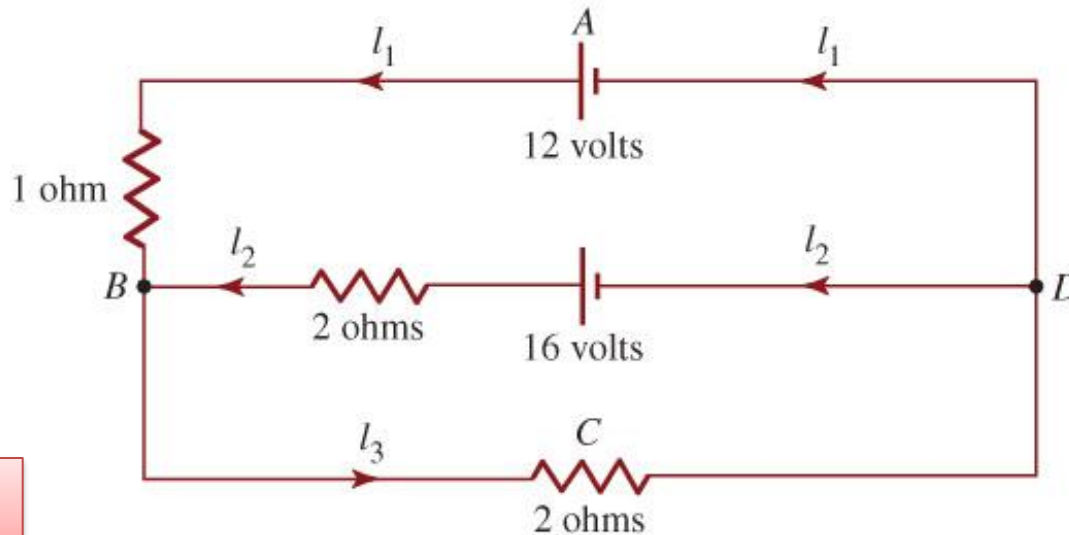


Figure 1.28

Solution

Junction

$$\text{Junction } B: I_1 + I_2 = I_3$$

$$\text{Junction } D: I_3 = I_1 + I_2$$

$$\text{giving } I_1 + I_2 - I_3 = 0.$$

Path

$$\text{Path } ABCDA: 1I_1 + 2I_3 = 12$$

$$\text{Path } ABDA: 1I_1 + 2(-I_2) = 12 + (-16)$$

We have three equations in the three variables, I_1 , I_2 , and I_3 .

$$\begin{aligned} I_1 + I_2 - I_3 &= 0 \\ I_1 + 2I_3 &= 12 \\ I_1 - 2I_2 &= -4 \end{aligned}$$

Solving these equations, we get $I_1 = 2$, $I_2 = 3$, $I_3 = 5$ (amps).

Traffic Flow

All traffic entering an intersection must leave that intersection.

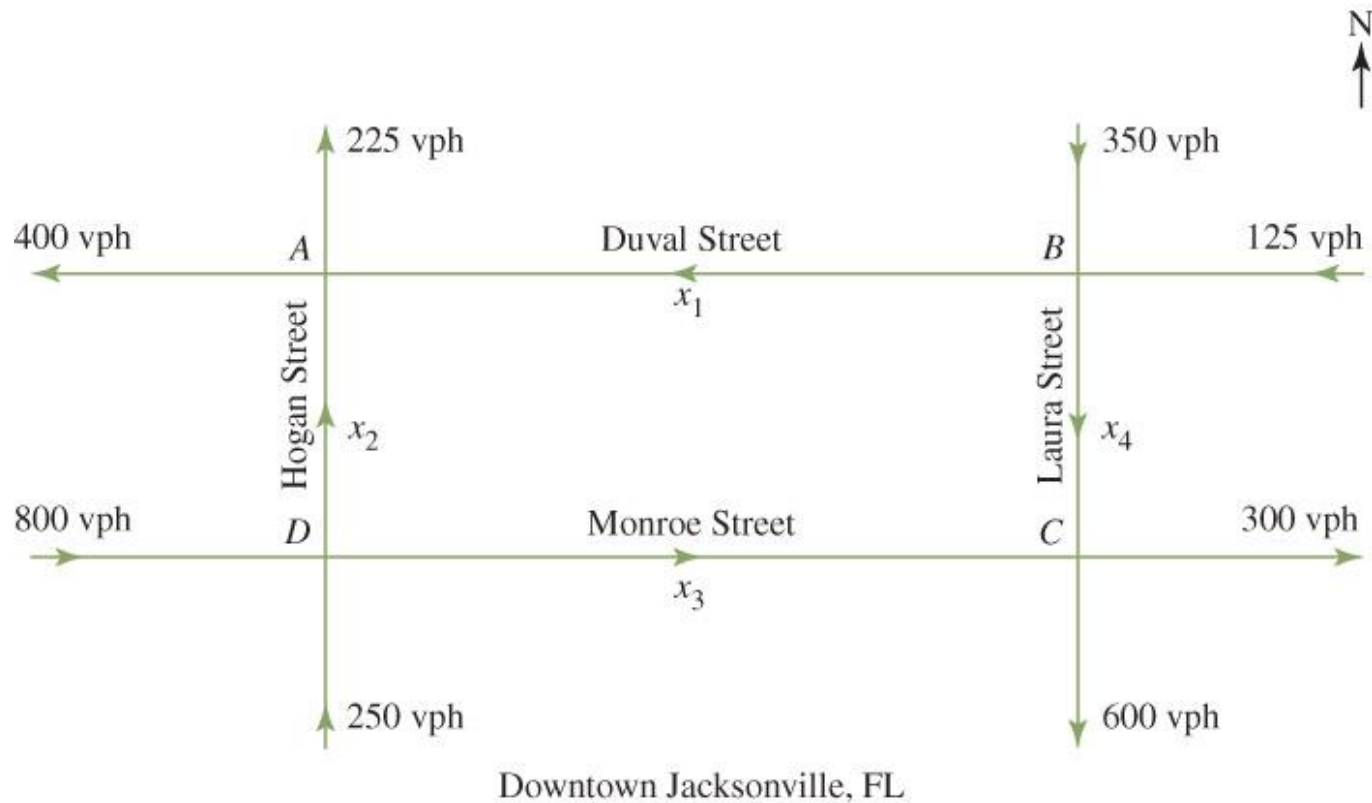


Figure 1.29

A : Traffic in = $x_1 + x_2$. Traffic out = $400 + 225$. Thus $x_1 + x_2 = 625$.

B : Traffic in = $350 + 125$. Traffic out = $x_1 + x_4$. Thus $x_1 + x_4 = 475$.

C : Traffic in = $x_3 + x_4$. Traffic out = $600 + 300$. Thus $x_3 + x_4 = 900$.

D : Traffic in = $800 + 250$. Traffic out = $x_2 + x_3$. Thus $x_2 + x_3 = 1050$.

The constraints on the traffic are described by the following system of linear equations.

$$x_1 + x_2 = 625$$

$$x_1 + x_4 = 475$$

$$x_3 + x_4 = 900$$

$$x_2 + x_3 = 1050$$

The method of Gauss-Jordan elimination is used to solve this system of equations. The augmented matrix and reduced echelon form of the preceding system are as follows:

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 625 \\ 1 & 0 & 0 & 1 & 475 \\ 0 & 0 & 1 & 1 & 900 \\ 0 & 1 & 1 & 0 & 1050 \end{bmatrix} \approx \dots \approx \begin{bmatrix} 1 & 0 & 0 & 1 & 475 \\ 0 & 1 & 0 & -1 & 150 \\ 0 & 0 & 1 & 1 & 900 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The system of equations that corresponds to this reduced echelon form is

$$x_1 + x_4 = 475$$

$$x_2 - x_4 = 150$$

$$x_3 + x_4 = 900$$

Expressing each leading variable in terms of the remaining variable, we get

$$x_1 = -x_4 + 475$$

$$x_2 = x_4 + 150$$

$$x_3 = -x_4 + 900$$

As was perhaps to be expected the system of equations has many solutions — there are many traffic flows possible. One does have a certain amount of choice at intersections.

Let us now use this mathematical model to arrive at information. Suppose it becomes necessary to perform road work on the stretch *DC* of Monroe Street. It is desirable to have as small a flow x_3 as possible along this stretch of road. The flows can be controlled along various branches by means of traffic lights. What is the minimum value of x_3 along *DC* that would not lead to traffic congestion? We use the preceding system of equations to answer this question.